Motion control: gain estimation

1 Discrete-time motion model and control law

Normalized variables

$$\bar{x} = \frac{x}{R}, \qquad \bar{a} = \frac{a_x}{a_{\text{nom}}}, \qquad \bar{f} = a_o\, f$$

$$a_{\text{nom}} = \frac{\Delta t}{\gamma_N} \;\; [\mu\text{m/pN}], \qquad a_o = \frac{a_{\text{nom}}}{R} \;\; [1/\text{pN}]$$

$$\Delta x = a_x\, f \quad \Longrightarrow \quad \Delta\bar{x} = \frac{a_{\text{nom}}}{R}\,\bar{a}\, f = a_o\,\bar{a}\, f = \bar{a}\,\bar{f}$$

Assumptions

wall surface at $\bar{x} = 0$ (position known)

no disturbed motion

each axis: the same one-dimensional equations with its own $\bar{a}$ $\;(x, y \parallel \text{wall},\; z \perp \text{wall})$

Motion model

$$\bar{x}[k+1] = \bar{x}[k] + \bar{a}[k]\,\big(\bar{f}_d[k] + \bar{f}_T[k]\big)$$

Tracking error, control objective, measured tracking error

$$\delta\bar{x}[k] = \bar{x}_d[k] - \bar{x}[k]$$

$$\delta\bar{x}[k+1] = \lambda_c\,\delta\bar{x}[k]$$

$$\bar{x}_m[k] = \bar{x}[k-d] + n_{\bar{x}}[k]$$

$$\delta\bar{x}_m[k] = \bar{x}_d[k-d] - \bar{x}_m[k]$$

Motion control law

$$\hat{\bar{x}}[k] = \bar{x}_m[k] + \sum_{i=1}^{d} \hat{\bar{a}}[k-i]\,\bar{f}_d[k-i]$$

$$\bar{f}_d[k] = \hat{\bar{a}}^{-1}[k]\,\Big\{\bar{x}_d[k+1] - \hat{\bar{x}}[k] - \lambda_c\,\big(\bar{x}_d[k] - \hat{\bar{x}}[k]\big)\Big\}$$

Dynamics of tracking error

$$\delta\bar{x}[k+1] = \lambda_c\,\delta\bar{x}[k] - \varepsilon[k]$$

$$\varepsilon[k] = e_{\bar{a}}[k]\,\bar{f}_d[k] + (1-\lambda_c)\sum_{i=1}^{d} e_{\bar{a}}[k-i]\,\bar{f}_d[k-i] + \varepsilon_x[k]$$

$$\varepsilon_x[k] = \bar{a}[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{a}[k-i]\,\bar{f}_T[k-i] - (1-\lambda_c)\,n_{\bar{x}}[k]$$

$$e_{\bar{a}}[k] = \bar{a}[k] - \hat{\bar{a}}[k]$$

$e_{\bar{a}}$ constant over d steps

$$\varepsilon[k] = e_{\bar{a}}[k]\, F_{dx}[k] + \varepsilon_x[k], \qquad F_{dx}[k] = \bar{f}_d[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{f}_d[k-i]$$

2 Effect of the motion gain on tracking

Control law with the nominal motion gain

$$\hat{\bar{a}}[k] = 1 \quad \Longrightarrow \quad \varepsilon[k] = \big(\bar{a}[k] - 1\big)\, F_{dx}[k] + \varepsilon_x[k]$$

Control law with the true motion gain

$$\hat{\bar{a}}[k] = \bar{a}[k] \quad \Longrightarrow \quad \varepsilon[k] = \varepsilon_x[k]$$

Desired trajectory

$$x_d(t) = y_d(t) = 9 \cdot \frac{2}{\pi} \arcsin\big(\sin(2\pi t)\big) \;\; \mu\text{m}$$

$$z_d(t) = 50 - 9.505\, t \;\; \mu\text{m}, \qquad 0 \le t \le 5 \text{ s} \qquad \big(z_d/R:\; 22.2 \to 1.10\big)$$

Fig. 2.1 Desired trajectory

x, y: 1 Hz triangles, amplitude 9 μm; z: distance from the wall, constant velocity

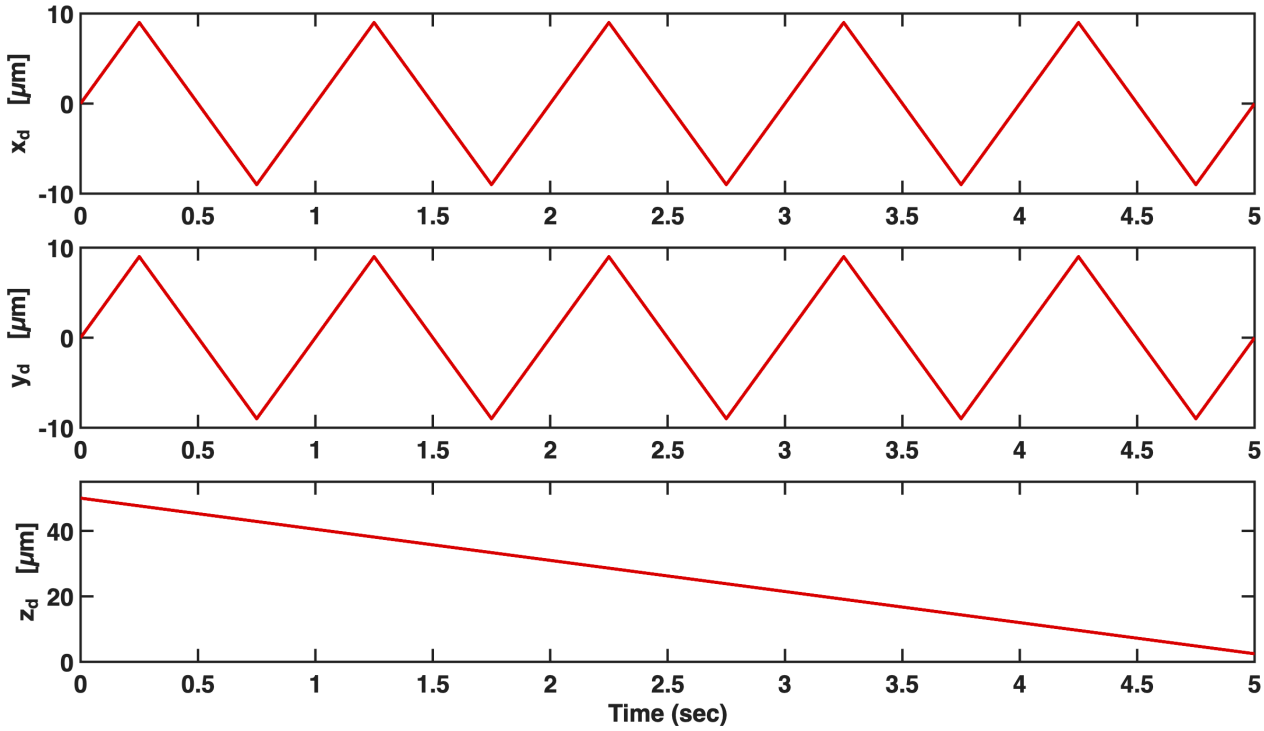
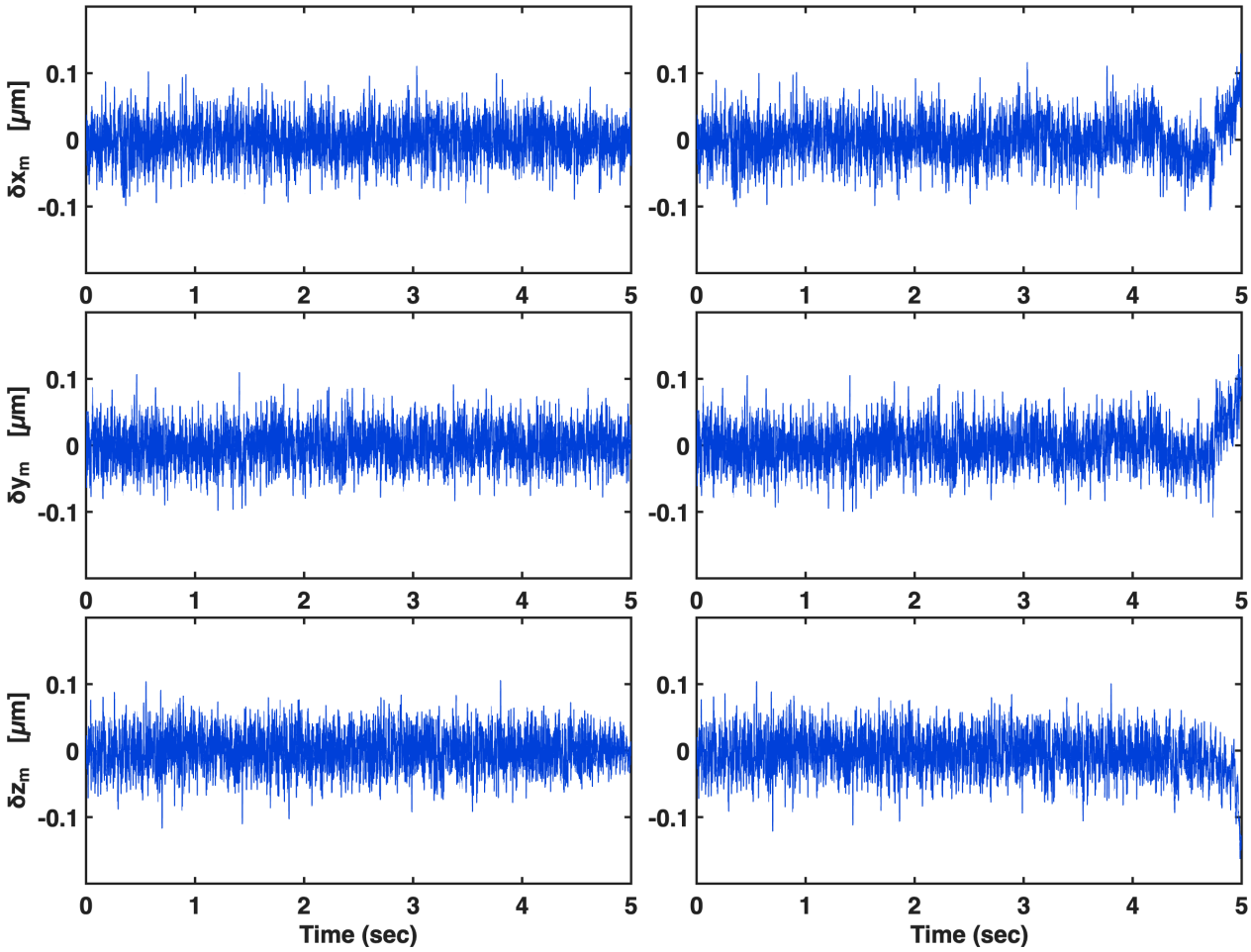


Fig. 2.2 Measured tracking error

left: $\hat{\bar{a}} = \bar{a}$ (true motion gain); right: $\hat{\bar{a}} = 1$ (nominal motion gain); plane wall, $\lambda_c = 0.4$, seed 1



3 Real-time calculation of process outputs

Deterministic and random components of the measured tracking error

$$\overline{\delta\bar{x}_m}[k] = a_{\mathrm{pd}} \cdot \delta\bar{x}_m[k] + (1 - a_{\mathrm{pd}})\, \overline{\delta\bar{x}_m}[k-1]$$
$$\delta\bar{x}_r[k] = \delta\bar{x}_m[k] - \overline{\delta\bar{x}_m}[k]$$

Variance of the random component

$$\hat{\sigma}^2_{\delta\bar{x}_r}[k] = a_{\mathrm{cov}} \cdot \delta\bar{x}_r^2[k] + (1 - a_{\mathrm{cov}})\, \hat{\sigma}^2_{\delta\bar{x}_r}[k-1]$$
$$\sigma^2_{\delta\bar{x}_T}[k] = \kappa_T\, \bar{a}[k], \qquad \kappa_T = \frac{4k_BT\, a_o}{R}$$
$$\sigma^2_{\delta\bar{x}_r}[k] = C_T\, \sigma^2_{\delta\bar{x}_T}[k] + C_n\, \sigma_n^2$$
$$\sigma^2_{\delta\bar{x}_r}[k] = C_T\, \kappa_T \left(\bar{a}[k] + \xi\right), \qquad \xi = \frac{C_n}{C_T}\, \frac{\sigma_n^2}{\kappa_T}$$

Coefficients

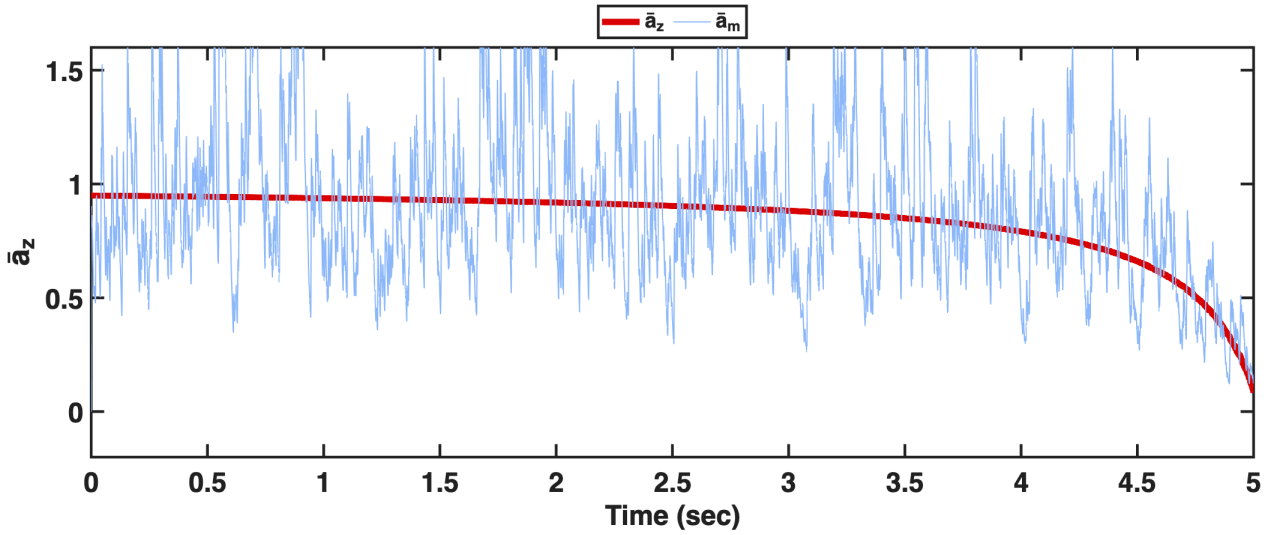
$$C_T = (1 - a_{\mathrm{pd}})^2 \left[\frac{2\,(1 - a_{\mathrm{pd}})(1 - \lambda_c)}{1 - (1 - a_{\mathrm{pd}})\lambda_c} + \frac{2}{(2 - a_{\mathrm{pd}})(1 + \lambda_c)\big(1 - (1 - a_{\mathrm{pd}})\lambda_c\big)} \right]$$
$$C_n = \frac{2\,(1 - a_{\mathrm{pd}})^2}{2 - a_{\mathrm{pd}}} \left[1 + \frac{(1 - a_{\mathrm{pd}})^2\, a_{\mathrm{pd}}\,(1 - \lambda_c)}{1 - (1 - a_{\mathrm{pd}})\lambda_c} + \frac{(1 - \lambda_c)^2}{(1 + \lambda_c)\big(1 - (1 - a_{\mathrm{pd}})\lambda_c\big)} \right]$$

Measured input gain

$$\bar{a}_m[k] = \frac{\hat{\sigma}^2_{\delta\bar{x}_r}[k] - C_n\, \sigma_n^2}{C_T\, \kappa_T}$$

Fig. 3.1 Measured input gain

z axis; Section 2 run with $\hat{\bar{a}} = \bar{a}$; plane wall, $\lambda_c = 0.4$, seed 1



4 Real-time estimation of motion gain and motion error: first-order process

State of the process

$$\delta\bar{x}_1[k] = \delta\bar{x}[k-2], \qquad \delta\bar{x}_2[k] = \delta\bar{x}[k-1], \qquad \delta\bar{x}_3[k] = \delta\bar{x}[k] \qquad (d=2)$$

$$\delta\bar{x}_1[k+1] = \delta\bar{x}_2[k]$$

$$\delta\bar{x}_2[k+1] = \delta\bar{x}_3[k]$$

$$\delta\bar{x}_3[k+1] = \lambda_c\,\delta\bar{x}_3[k] - F_{dx}[k]\,e_{\bar{a}}[k] - \varepsilon_x[k]$$

$$\bar{a}[k+1] = \bar{a}[k] + w_{\bar{a}}[k]$$

Measurement and output matrix

$$\mathbf{y}[k] = \begin{bmatrix} \delta\bar{x}_m[k] \\ \bar{a}_m[k] \end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix} \delta\hat{\bar{x}}_1[k] \\ \hat{\bar{a}}[k] \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Estimator

$$e_{x_1}[k] = \delta\bar{x}_m[k] - \delta\hat{\bar{x}}_1[k]$$

$$e_{a_m}[k] = \bar{a}_m[k] - \hat{\bar{a}}[k]$$

$$\delta\hat{\bar{x}}_1[k+1] = \delta\hat{\bar{x}}_2[k] + \left\{\ell_{11}[k]\,e_{x_1}[k] + \ell_{12}[k]\,e_{a_m}[k]\right\}$$

$$\delta\hat{\bar{x}}_2[k+1] = \delta\hat{\bar{x}}_3[k] + \left\{\ell_{21}[k]\,e_{x_1}[k] + \ell_{22}[k]\,e_{a_m}[k]\right\}$$

$$\delta\hat{\bar{x}}_3[k+1] = \lambda_c\,\delta\hat{\bar{x}}_3[k] + \left\{\ell_{31}[k]\,e_{x_1}[k] + \ell_{32}[k]\,e_{a_m}[k]\right\}$$

$$\hat{\bar{a}}[k+1] = \hat{\bar{a}}[k] + \left\{\ell_{41}[k]\,e_{x_1}[k] + \ell_{42}[k]\,e_{a_m}[k]\right\}$$

Dynamics of estimation errors

$$\mathbf{e}[k] = \begin{bmatrix} e_{\delta\bar{x}_1}[k] \\ e_{\delta\bar{x}_2}[k] \\ e_{\delta\bar{x}_3}[k] \\ e_{\bar{a}}[k] \end{bmatrix} = \begin{bmatrix} \delta\bar{x}_1[k] - \delta\hat{\bar{x}}_1[k] \\ \delta\bar{x}_2[k] - \delta\hat{\bar{x}}_2[k] \\ \delta\bar{x}_3[k] - \delta\hat{\bar{x}}_3[k] \\ \bar{a}[k] - \hat{\bar{a}}[k] \end{bmatrix}$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{q}[k] = \begin{bmatrix} 0 \\ 0 \\ -\varepsilon_x[k] \\ w_{\bar{a}}[k] \end{bmatrix}$$

Estimator feedback matrix and error covariance

$$\mathbf{L}[k] = \mathbf{P}^f[k]\,\mathbf{H}^T\left(\mathbf{H}\,\mathbf{P}^f[k]\,\mathbf{H}^T + \mathbf{R}[k]\right)^{-1}$$

$$\mathbf{P}[k] = \left(\mathbf{I} - \mathbf{L}[k]\,\mathbf{H}\right)\mathbf{P}^f[k]$$

$$\mathbf{P}^f[k+1] = \mathbf{F}_e[k]\,\mathbf{P}[k]\,\mathbf{F}_e^T[k] + \mathbf{Q}[k]$$

Process noise and measurement noise

$$\mathbf{Q}[k] = \mathrm{diag}\big(0,\ 0,\ q_{33}[k],\ q_{44}[k]\big)$$

$$q_{33}[k] = \kappa_T \left(\hat{\bar{a}}[k] + (1-\lambda_c)^2 \sum_{i=1}^{d} \hat{\bar{a}}[k-i]\right) + (1-\lambda_c)^2\, \sigma_n^2$$

$$q_{44}[k] = \varepsilon\, r_2[k]$$

$$\mathbf{R}[k] = \mathrm{diag}\big(r_1,\ r_2[k]\big)$$

$$r_1 = \sigma_n^2$$

$$r_2[k] = K_{\mathrm{var}}\, \mathrm{IF}[k]\, \big(\hat{\bar{a}}[k] + \xi\big)^2, \qquad K_{\mathrm{var}} = \frac{2\, a_{\mathrm{cov}}}{2 - a_{\mathrm{cov}}}$$

Fig. 4.1 Process noise of the motion gain: $\varepsilon = 0$ vs 10^{-5}

z axis; Section 2 trajectory; plane wall, $\lambda_c = 0.4$, seed 1; $q_{44} = \varepsilon r_2$

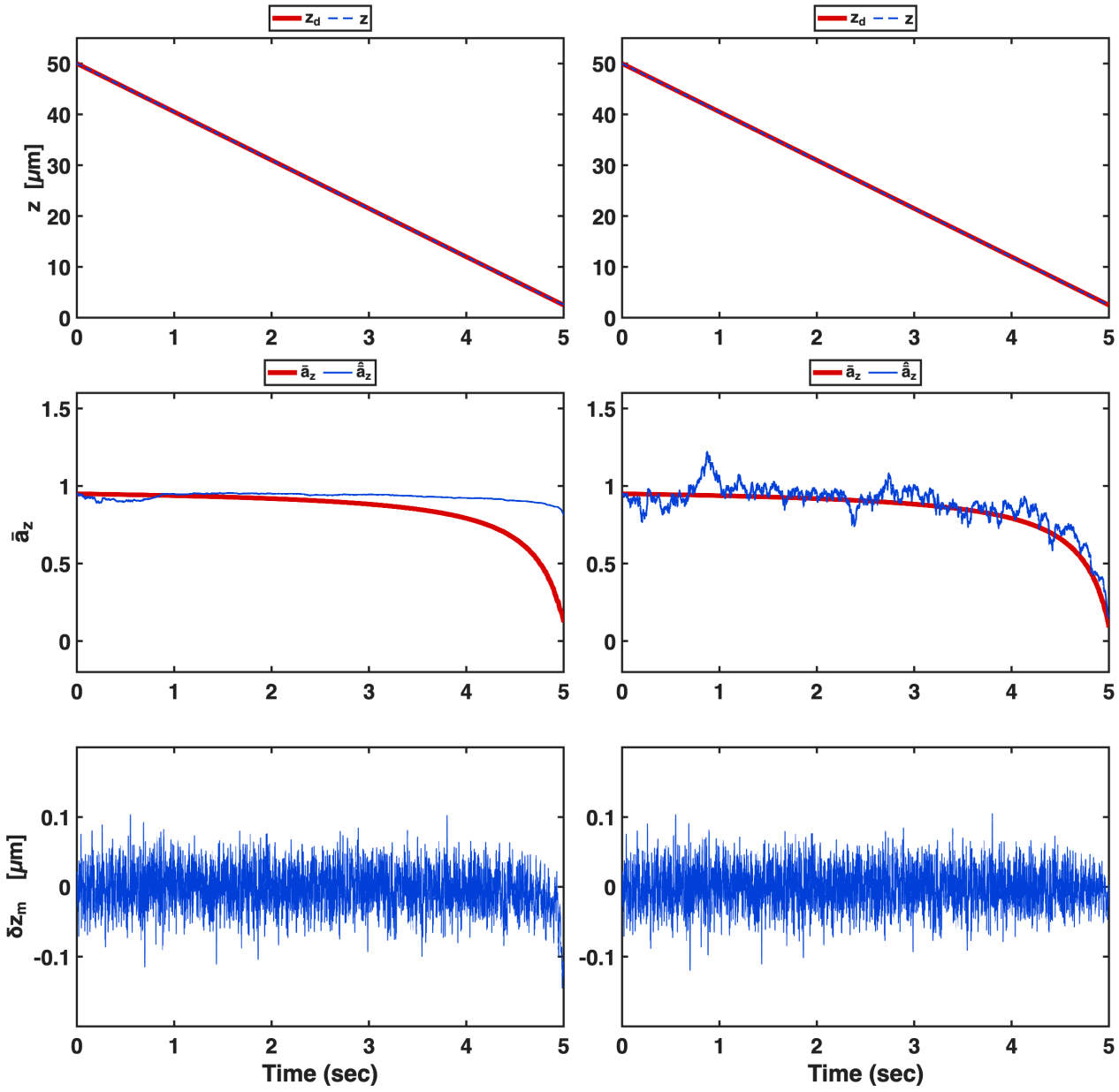


Fig. 4.2 Measured tracking error: nominal vs estimated motion gain

Section 2 trajectory, three axes; left: $\hat{\bar{a}} = 1$; right: first-order estimator, $\varepsilon = 10^{-5}$; plane wall, $\lambda_c = 0.4$, seed 1

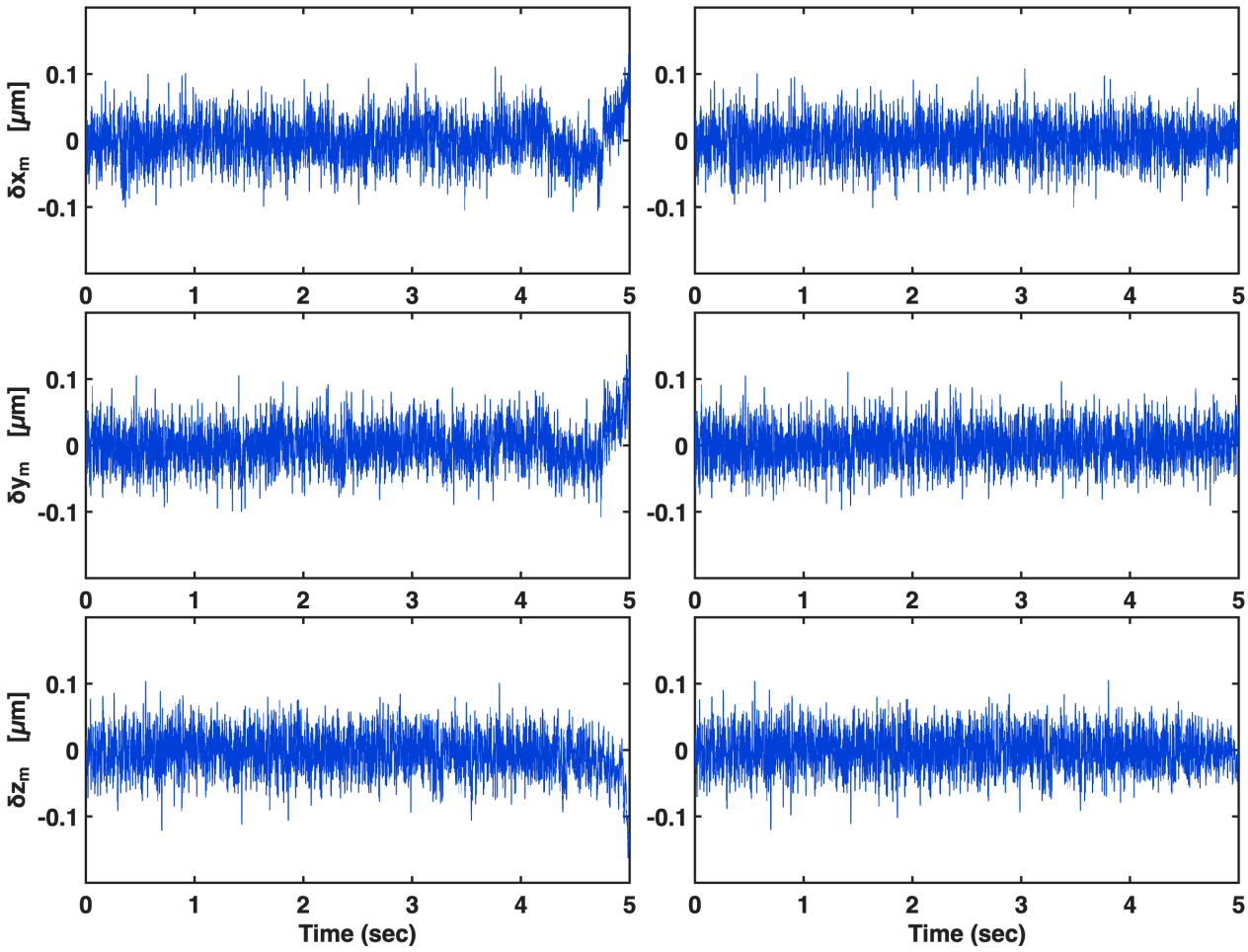
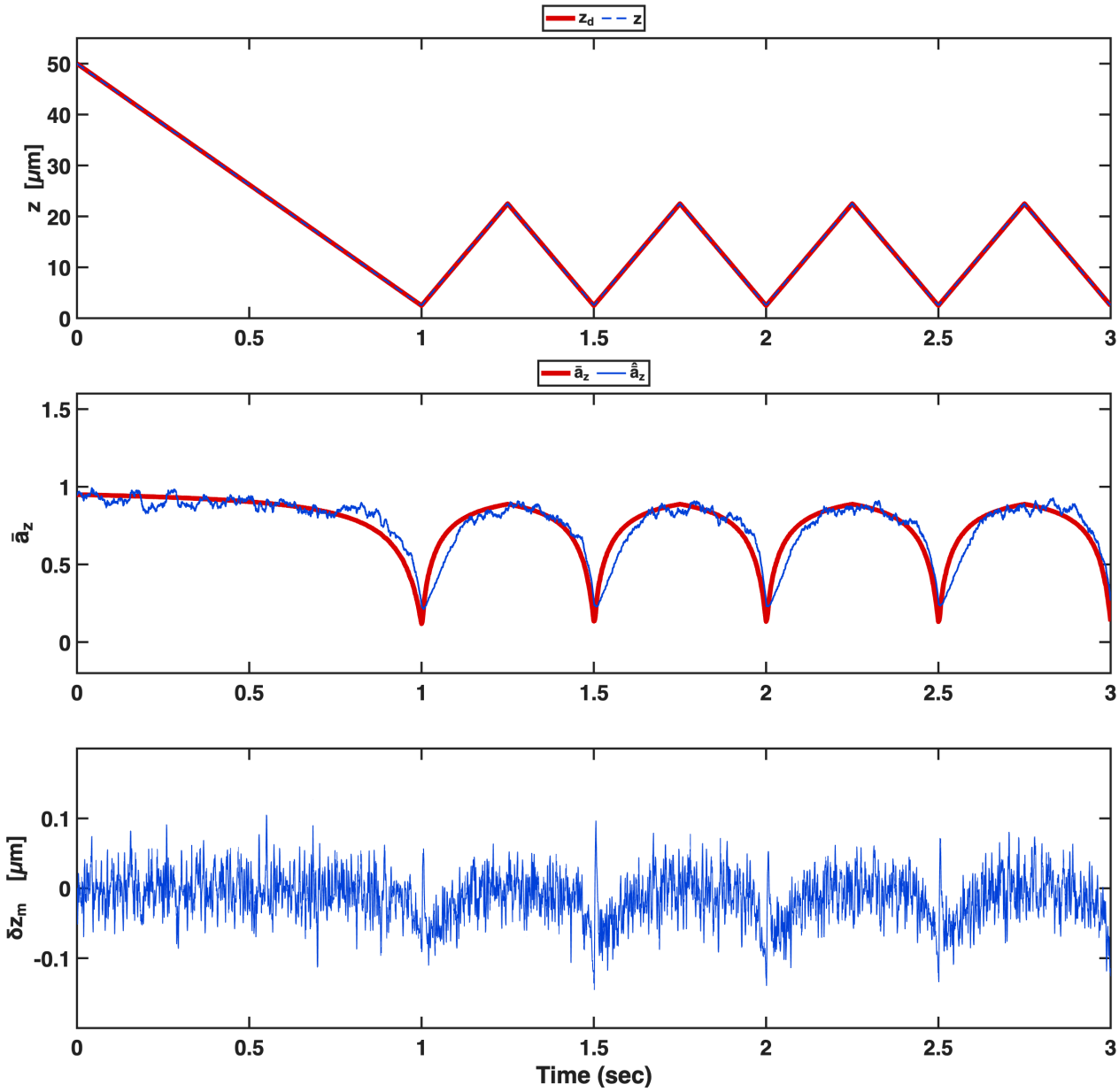


Fig. 4.3 First-order estimator on an oscillating trajectory

z axis; top: z_d (solid) and z (dashed), 22.2 → 1.10 R in 1 s, then triangle 1.10 ↔ 10 R at 2 Hz for 2 s; plane wall, $\lambda_c = 0.4$, $\varepsilon = 10^{-5}$, seed 1



5 Real-time estimation with a position-dependent motion gain

Motion gain as a function of the distance from the wall

$$
\begin{aligned}
h[k] &= \bar{x}_d[k] - \delta\bar{x}_3[k] \\
\bar{a}'[k] &= \frac{1-c}{h[k]^2} \\
M[k] &= \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\bar{x}_3[k]
\end{aligned}
$$

State of the process

$$
\begin{aligned}
\delta\bar{x}_1[k+1] &= \delta\bar{x}_2[k] \\
\delta\bar{x}_2[k+1] &= \delta\bar{x}_3[k] \\
\delta\bar{x}_3[k+1] &= \lambda_c\,\delta\bar{x}_3[k] - F_{dx}[k]\,e_{\bar{a}}[k] - \varepsilon_x[k] \\
\bar{a}[k+1] &= \bar{a}[k] + \bar{a}'[k]\left(M[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\right) \\
c[k+1] &= c[k]
\end{aligned}
$$

Measurement and output matrix

$$
\mathbf{y}[k] = \begin{bmatrix} \delta\bar{x}_m[k] \\ \bar{a}_m[k] \end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix} \delta\hat{\bar{x}}_1[k] \\ \hat{\bar{a}}[k] \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}
$$

Estimator

$$
\begin{aligned}
\hat{h}[k] &= \bar{x}_d[k] - \delta\hat{\bar{x}}_3[k] \\
\hat{\bar{a}}'[k] &= \frac{1-\hat{c}[k]}{\hat{h}[k]^2} \\
\hat{M}[k] &= \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{x}}_3[k]
\end{aligned}
$$

$$
\begin{aligned}
\delta\hat{\bar{x}}_1[k+1] &= \delta\hat{\bar{x}}_2[k] + \left\{\ell_{11}[k]\,e_{x_1}[k] + \ell_{12}[k]\,e_{a_m}[k]\right\} \\
\delta\hat{\bar{x}}_2[k+1] &= \delta\hat{\bar{x}}_3[k] + \left\{\ell_{21}[k]\,e_{x_1}[k] + \ell_{22}[k]\,e_{a_m}[k]\right\} \\
\delta\hat{\bar{x}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{x}}_3[k] + \left\{\ell_{31}[k]\,e_{x_1}[k] + \ell_{32}[k]\,e_{a_m}[k]\right\} \\
\hat{\bar{a}}[k+1] &= \hat{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\hat{M}[k] + \left\{\ell_{41}[k]\,e_{x_1}[k] + \ell_{42}[k]\,e_{a_m}[k]\right\} \\
\hat{c}[k+1] &= \hat{c}[k] + \left\{\ell_{51}[k]\,e_{x_1}[k] + \ell_{52}[k]\,e_{a_m}[k]\right\}
\end{aligned}
$$

Dynamics of estimation errors

$$
\mathbf{e}[k] = \begin{bmatrix} e_{\delta\bar{x}_1}[k] \\ e_{\delta\bar{x}_2}[k] \\ e_{\delta\bar{x}_3}[k] \\ e_{\bar{a}}[k] \\ e_c[k] \end{bmatrix}, \qquad \mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k]
$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & 0 \\ 0 & 0 & \hat{\bar{a}}'[k]\,(1-\lambda_c) + \dfrac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k] & 1 + \hat{\bar{a}}'[k]\,F_{dx}[k] & -\dfrac{\hat{M}[k]}{\hat{h}[k]^2} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{q}[k] = \begin{bmatrix} 0 \\ 0 \\ -\varepsilon_x[k] \\ \hat{\bar{a}}'[k]\,\varepsilon_x[k] \\ 0 \end{bmatrix}$$

Row 4 of $\mathbf{F}_e$, expanded

(a) the two gain steps and their difference

$$\bar{a}[k+1] - \bar{a}[k] = \bar{a}'[k]\,\big(M[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\big)$$
$$\hat{\bar{a}}[k+1] - \hat{\bar{a}}[k] = \hat{\bar{a}}'[k]\,\hat{M}[k]$$
$$e_{\bar{a}}[k+1] - e_{\bar{a}}[k] = \bar{a}'[k]\,M[k] - \hat{\bar{a}}'[k]\,\hat{M}[k] + \bar{a}'[k]\,F_{dx}[k]\,e_{\bar{a}}[k] + \bar{a}'[k]\,\varepsilon_x[k]$$

(b) each factor written as its estimate plus its error

$$h[k] = \hat{h}[k] - e_{\delta\bar{x}_3}[k], \qquad c = \hat{c}[k] + e_c[k]$$
$$\bar{a}'[k] = \frac{1 - \hat{c}[k] - e_c[k]}{\big(\hat{h}[k] - e_{\delta\bar{x}_3}[k]\big)^2} = \hat{\bar{a}}'[k] + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,e_{\delta\bar{x}_3}[k] - \frac{1}{\hat{h}[k]^2}\,e_c[k] + O(e^2)$$
$$M[k] = \hat{M}[k] + (1 - \lambda_c)\,e_{\delta\bar{x}_3}[k]$$

(c) the product difference, exact

$$\bar{a}'[k]\,M[k] - \hat{\bar{a}}'[k]\,\hat{M}[k] = \hat{\bar{a}}'[k]\,\big(M[k] - \hat{M}[k]\big) + \big(\bar{a}'[k] - \hat{\bar{a}}'[k]\big)\,\hat{M}[k] + \big(\bar{a}'[k] - \hat{\bar{a}}'[k]\big)\big(M[k] - \hat{M}[k]\big)$$

(d) every term after substituting (b)

$$\begin{aligned} e_{\bar{a}}[k+1] - e_{\bar{a}}[k] = {} & \hat{\bar{a}}'[k]\,(1 - \lambda_c)\,e_{\delta\bar{x}_3}[k] \\ & + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\delta\bar{x}_3}[k] - \frac{\hat{M}[k]}{\hat{h}[k]^2}\,e_c[k] \\ & + \hat{\bar{a}}'[k]\,F_{dx}[k]\,e_{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\varepsilon_x[k] \\ & + \Big(\frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,e_{\delta\bar{x}_3}[k] - \frac{e_c[k]}{\hat{h}[k]^2}\Big)\Big((1 - \lambda_c)\,e_{\delta\bar{x}_3}[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\Big) + O(e^2) \end{aligned}$$

(e) first-order terms collected by error

$$e_{\bar{a}}[k+1] = \Big(\hat{\bar{a}}'[k]\,(1 - \lambda_c) + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\Big)\,e_{\delta\bar{x}_3}[k] + \big(1 + \hat{\bar{a}}'[k]\,F_{dx}[k]\big)\,e_{\bar{a}}[k] - \frac{\hat{M}[k]}{\hat{h}[k]^2}\,e_c[k] + \hat{\bar{a}}'[k]\,\varepsilon_x[k]$$

Process noise and measurement noise

$$\mathbf{Q}[k] = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & q_{33}[k] & -\hat{\bar{a}}'[k]\,q_{33}[k] & 0 \\ 0 & 0 & -\hat{\bar{a}}'[k]\,q_{33}[k] & \hat{\bar{a}}'[k]^2\,q_{33}[k] & 0 \\ 0 & 0 & 0 & 0 & q_{55} \end{bmatrix}, \qquad q_{55} = 0$$

$$\mathbf{R}[k] = \mathrm{diag}\big(r_1,\ r_2[k]\big)$$

(q_{33}, r_1, r_2, $\mathbf{L}$, $\mathbf{P}$, $\mathbf{P}^f$ as in Section 4.)

Fig. 5.1 First order vs position-dependent motion gain: descent

z axis; Section 2 trajectory; left: Section 4 estimator; right: this section, $q_{55} = 0$; plane wall, $\lambda_c = 0.4$, seed 1

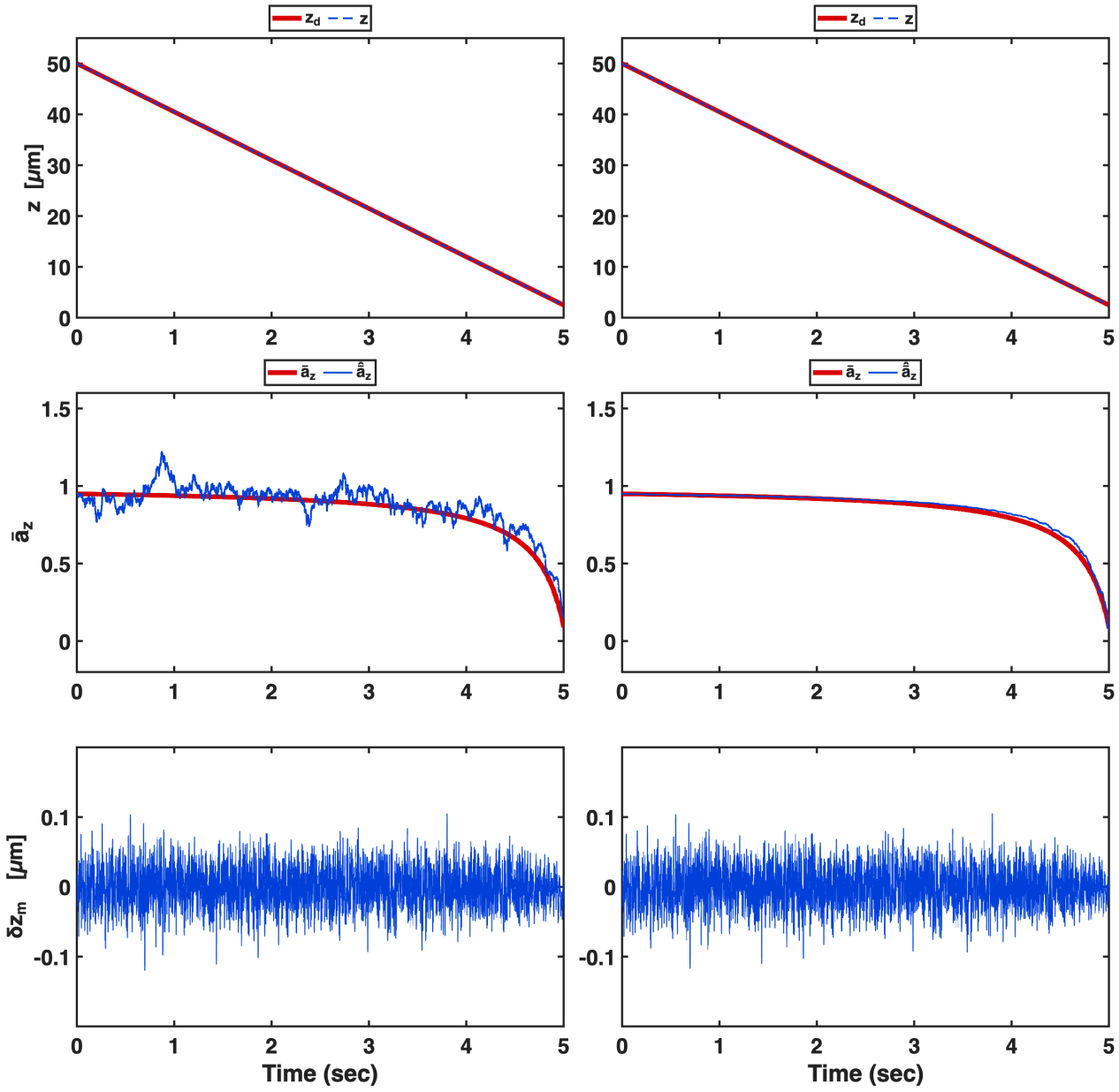


Fig. 5.2 Change of the motion-gain estimate in one step: prediction and correction

z axis; run of Fig. 5.1; left: Section 4; right: this section; top: accumulated from $t = 0$; bottom: $|\ell_{41}|$, $|\ell_{42}|$ (50 ms mean, log axis); seed 1

Section 4: $\hat{\bar{a}}[k+1] - \hat{\bar{a}}[k] = \ell_{41}[k]\, e_{x_1}[k] + \ell_{42}[k]\, e_{a_m}[k]$

this section: $\hat{\bar{a}}[k+1] - \hat{\bar{a}}[k] = \hat{\bar{a}}'[k]\, \hat{M}[k] + \ell_{41}[k]\, e_{x_1}[k] + \ell_{42}[k]\, e_{a_m}[k]$

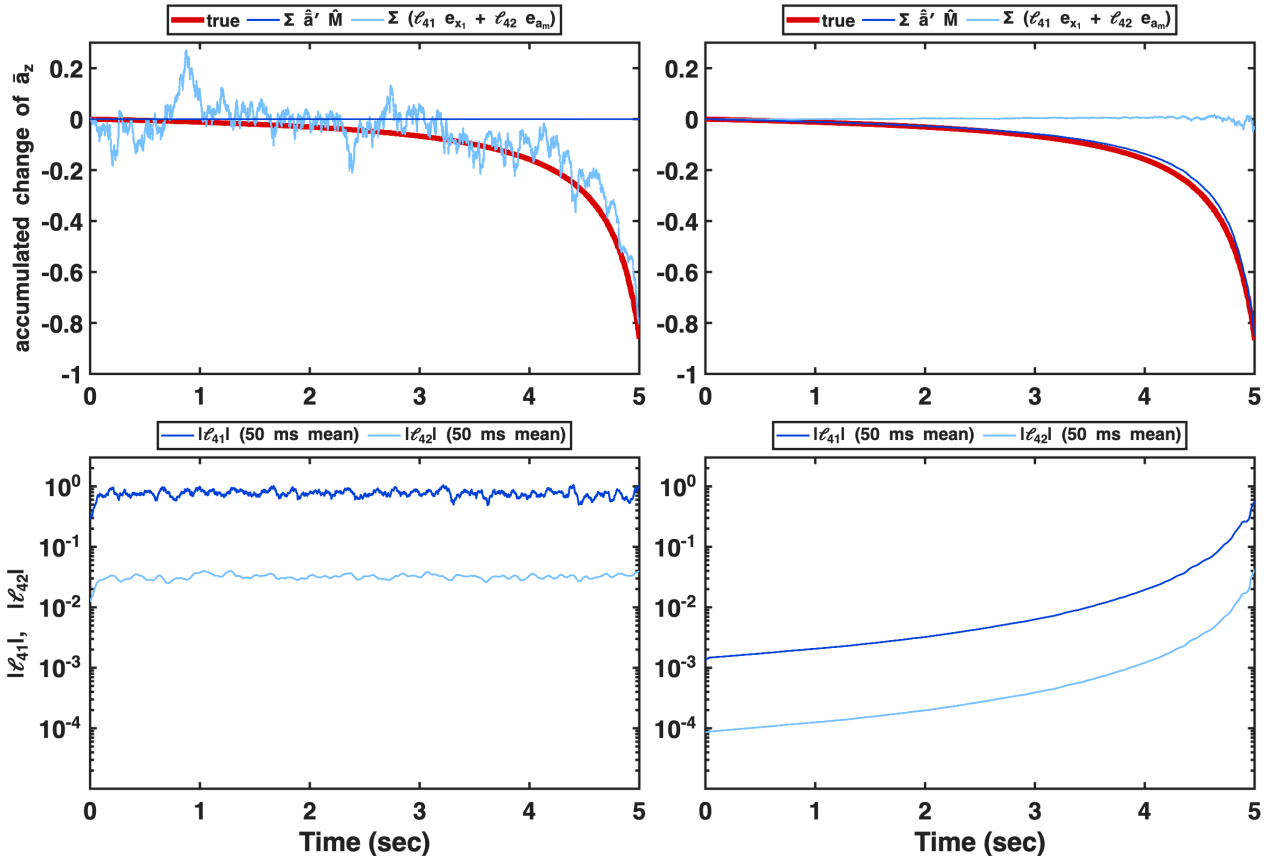
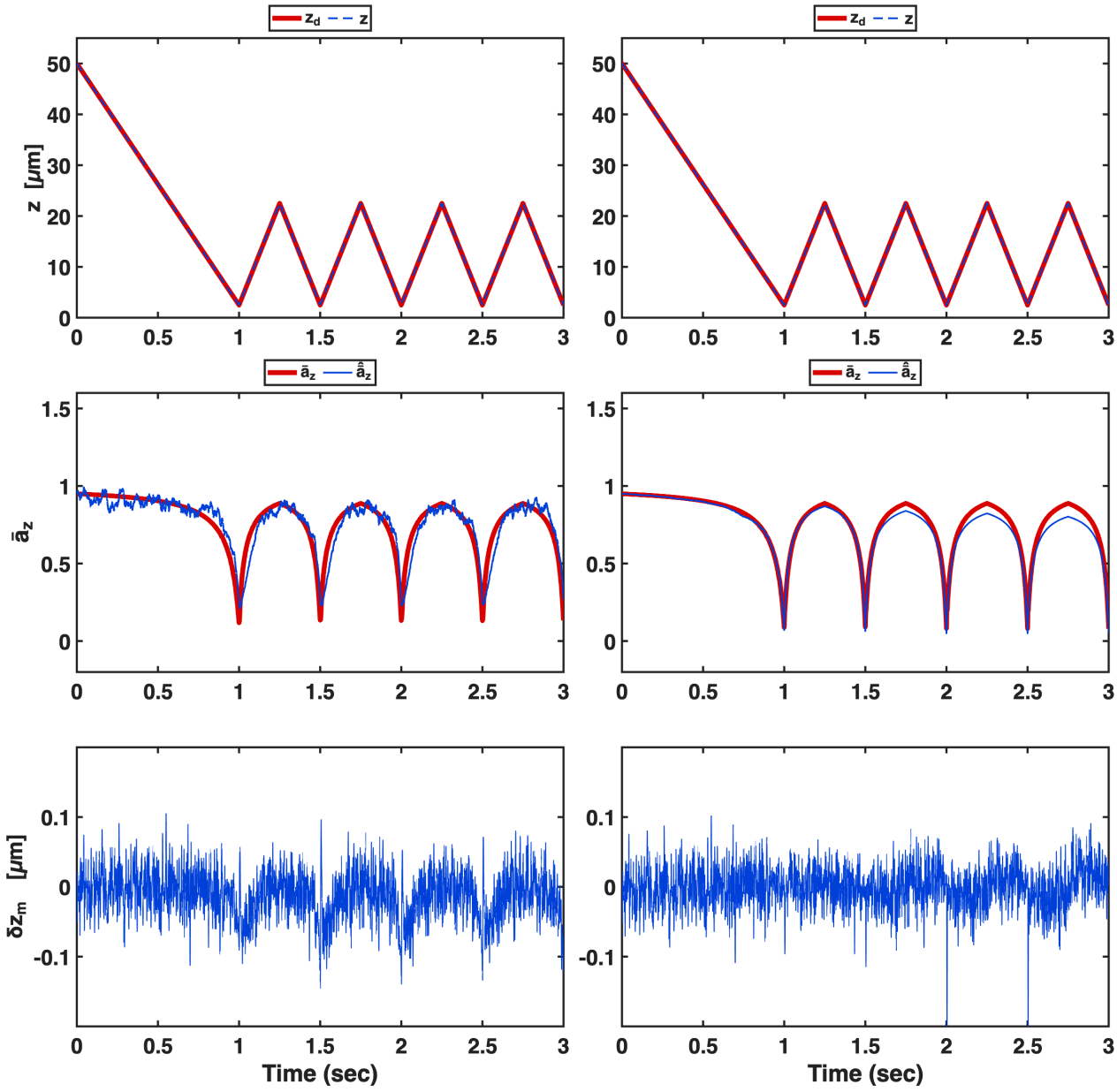


Fig. 5.3 First order vs position-dependent motion gain: oscillation

z axis; trajectory of Fig. 4.3 (triangle 1.10 ↔ 10 R, 2 Hz); left: Section 4 estimator; right: this section, $q_{55} = 0$; plane wall, $\lambda_c = 0.4$, seed 1



6 Unknown wall type: process noise of c

Motion gain of the two walls (plant only)

$$D_\perp(u) = 1 - \tfrac{9}{8}\,u + \tfrac{1}{2}\,u^3 - \tfrac{57}{100}\,u^4 + \tfrac{1}{5}\,u^5 + \tfrac{7}{200}\,u^{11} - \tfrac{1}{25}\,u^{12}$$

$$\bar{a}_{\text{plane}}(h) = D_\perp\Big(\frac{1}{h}\Big)$$

$$\bar{a}_{\text{cell}}(h) = D_\perp\Big(\frac{1}{h+1.05}\Big)$$

Value of c that makes the motion-gain slope exact at h

$$\bar{a}'(h) = \frac{1-c(h)}{h^2} \quad\Longrightarrow\quad c(h) = 1 - h^2\,\bar{a}'(h)$$

Feedback gain on c (row 5 of $\mathbf{L}[k]$, $\mathbf{H}$ of Section 5)

$$\big[\,\ell_{51}[k],\ \ell_{52}[k]\,\big] = \big[\,p^f_{51}[k],\ p^f_{54}[k]\,\big]\,\big(\mathbf{H}\,\mathbf{P}^f[k]\,\mathbf{H}^T + \mathbf{R}[k]\big)^{-1}$$

Update of $\hat{c}$ and of its variance

$$\hat{c}[k+1] = \hat{c}[k] + \ell_{51}[k]\,e_{x_1}[k] + \ell_{52}[k]\,e_{a_m}[k]$$

$$p_{55}[k] = p^f_{55}[k] - \ell_{51}[k]\,p^f_{15}[k] - \ell_{52}[k]\,p^f_{45}[k]$$

Information gained on c at step k

$$I[k] = \frac{1}{p_{55}[k]} - \frac{1}{p^f_{55}[k]} \;\geq\; 0$$

Forecast of the variance

$$p^f_{55}[k+1] = p_{55}[k] + q_{55}[k]$$

Choice of q_{55}: constant variance $p^f_{55} = P_{\text{fix}}$

$$p^f_{55}[k+1] = p^f_{55}[k] = P_{\text{fix}}$$

$$q_{55}[k] = P_{\text{fix}} - p_{55}[k]$$

$$\frac{1}{p_{55}[k]} = \frac{1}{P_{\text{fix}}} + I[k]$$

$$q_{55}[k] = \frac{P^2_{\text{fix}}\,I[k]}{1 + P_{\text{fix}}\,I[k]}$$

$$P_{\text{fix}}\,I[k] \ll 1 \quad\Longrightarrow\quad \boxed{q_{55}[k] = P^2_{\text{fix}}\,I[k]}$$

Fig. 6.1 Motion gain of the two walls

$\bar{a}_{\text{plane}}(h)$ and $\bar{a}_{\text{cell}}(h)$; h from 1.10 to 22.2 R

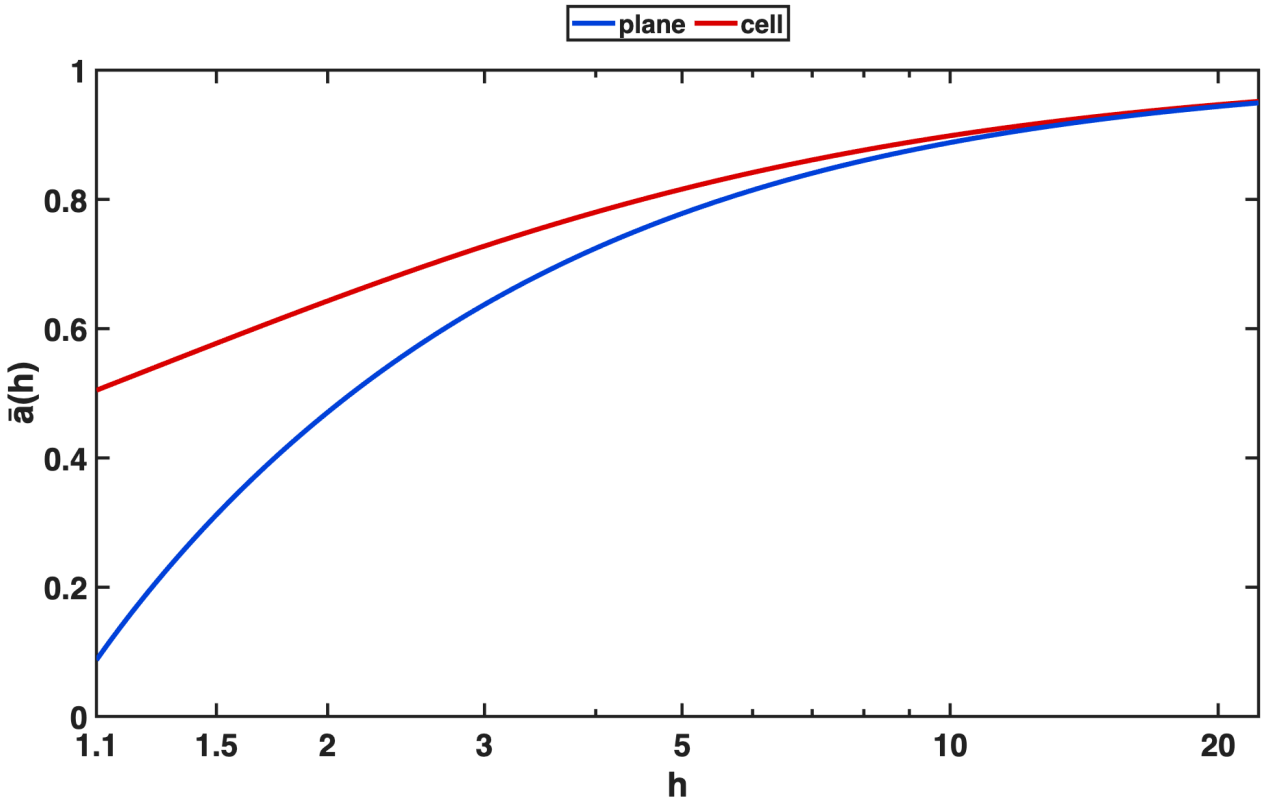


Fig. 6.2 Slope term of the two walls, $1 - c(h) = h^2\,\bar{a}'(h)$

plane and cell; h from 1.10 to 22.2 R; top: $1 - c(h)$; bottom: the smaller of the two divided by the larger

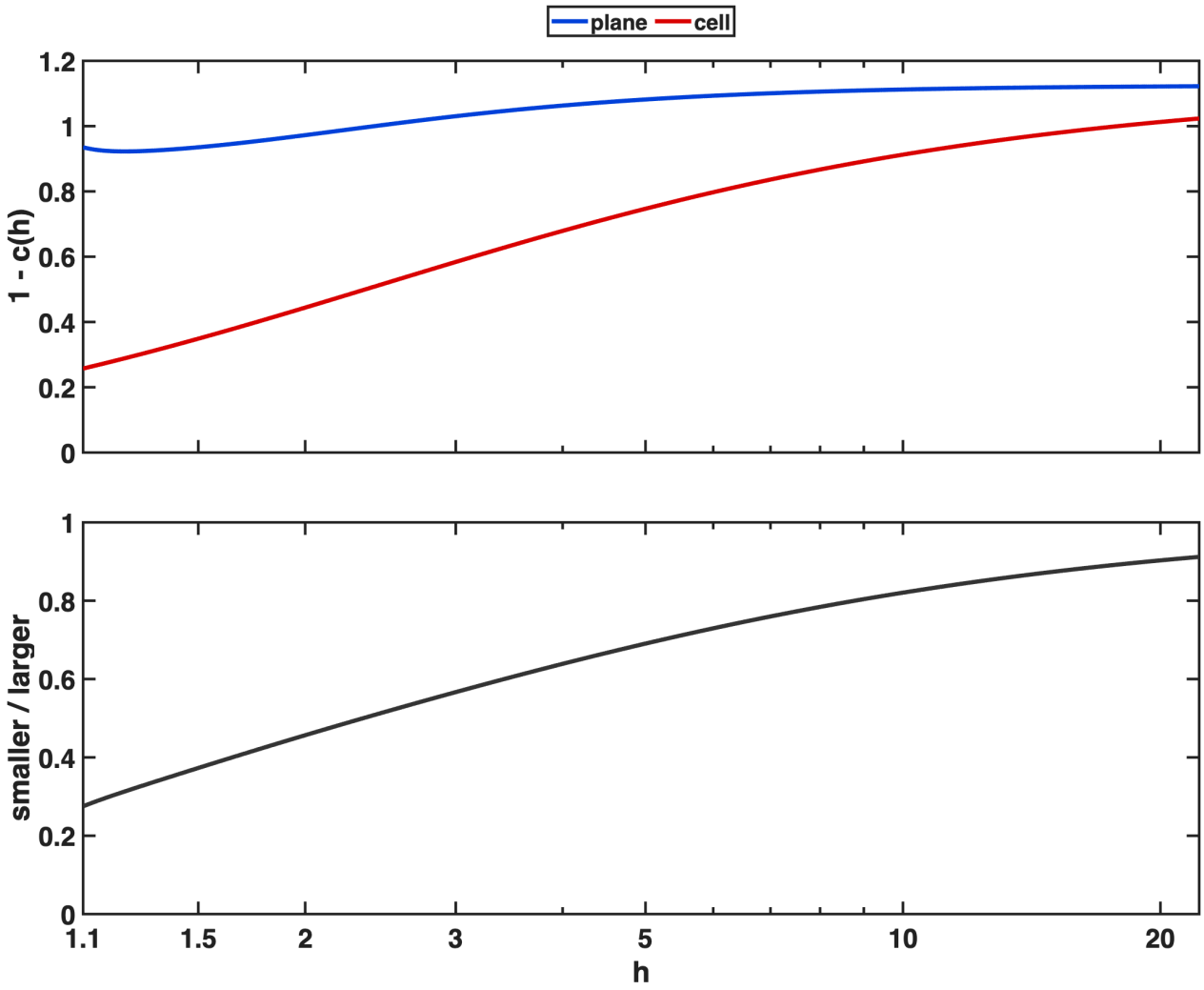


Fig. 6.3 Plane wall: $q_{55} = 0$ vs $q_{55} = P_{\text{fix}}^2 I$

z axis; trajectory of Fig. 4.3; left: $q_{55} = 0$ (Section 5); right: $\sqrt{P_{\text{fix}}} = 0.15$; $\lambda_c = 0.4$, seed 1

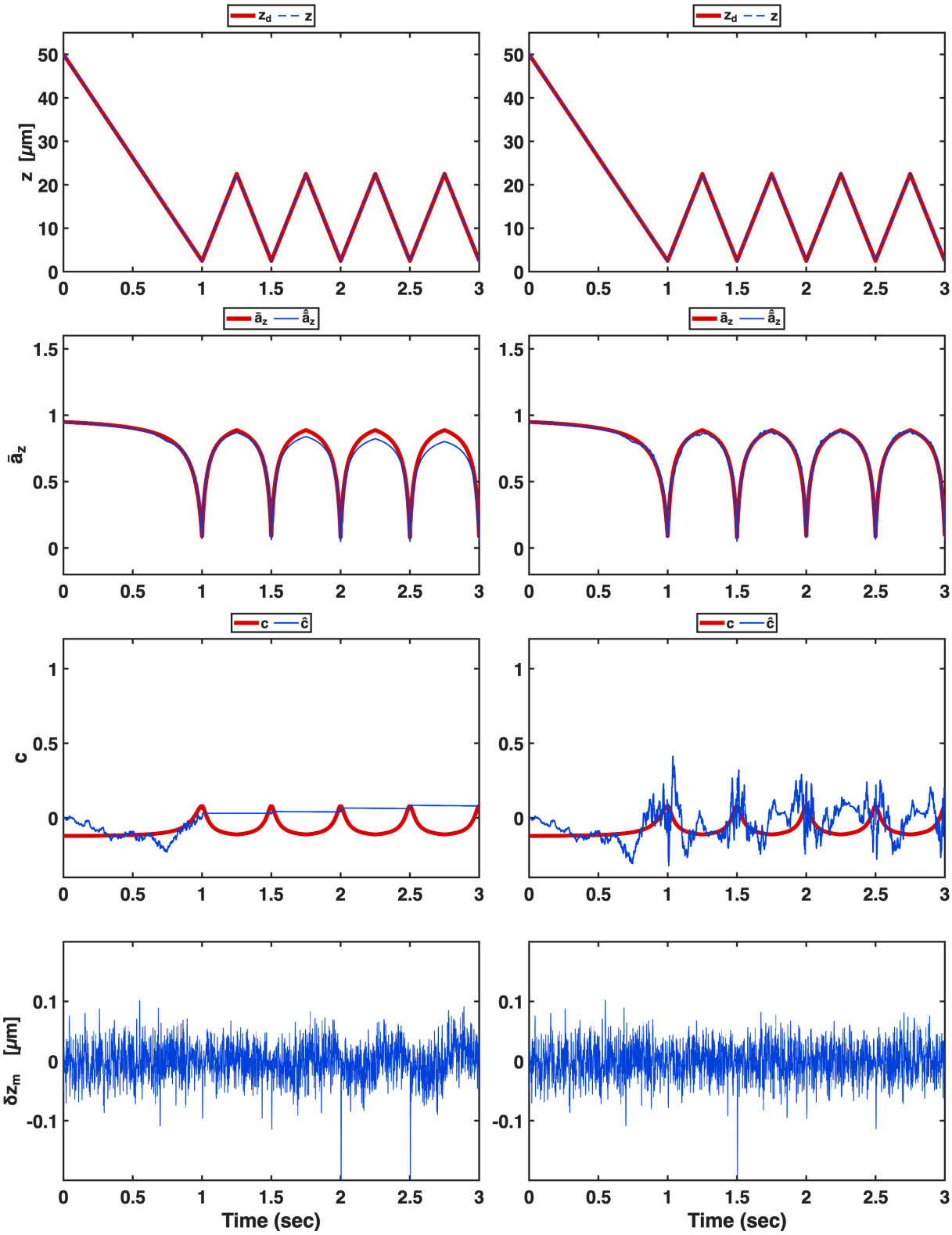


Fig. 6.4 Cell wall: $q_{55} = 0$ vs $q_{55} = P_{\text{fix}}^2 I$

z axis; trajectory of Fig. 4.3; left: $q_{55} = 0$; right: $\sqrt{P_{\text{fix}}} = 0.15$; $\lambda_c = 0.4$, seed 1

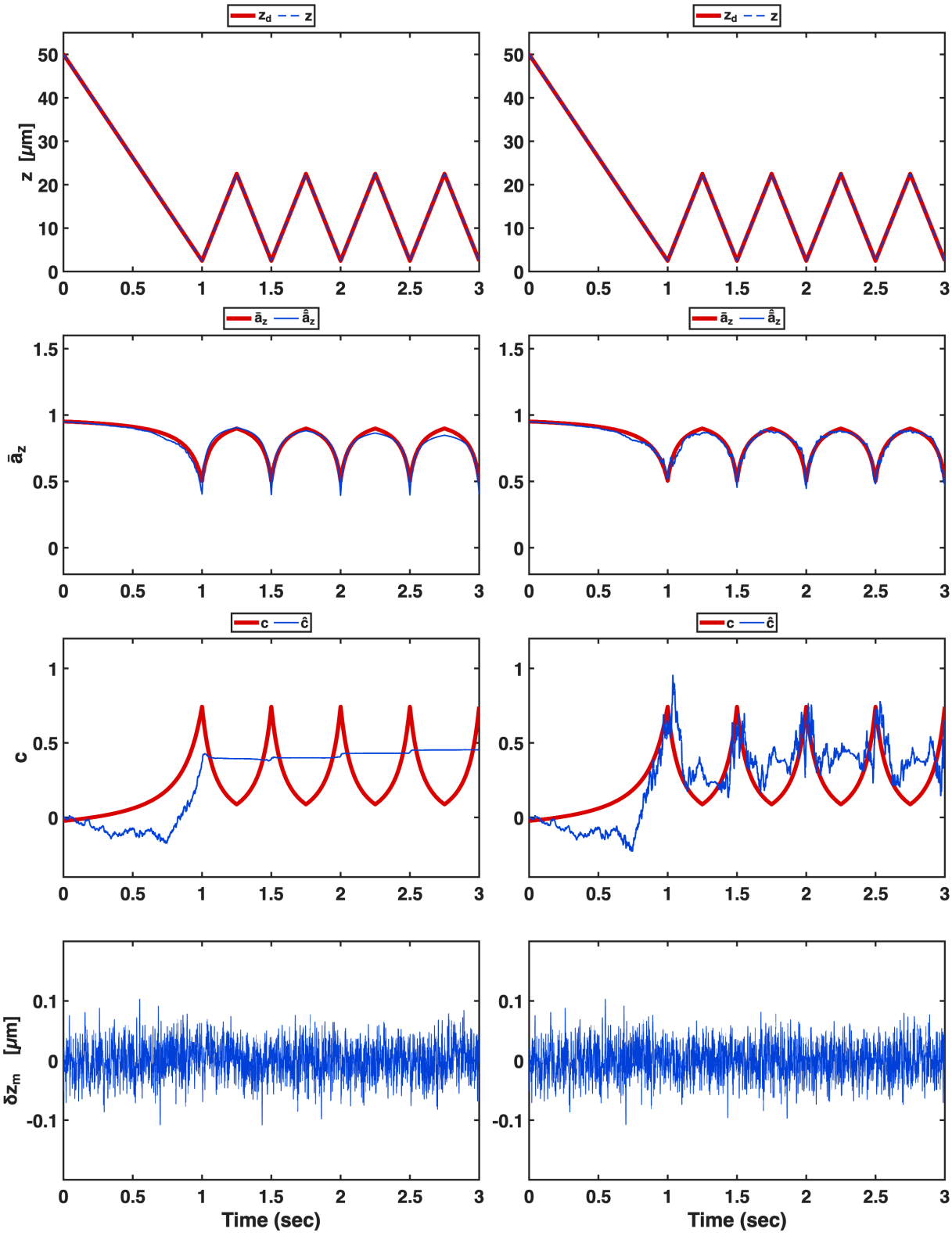
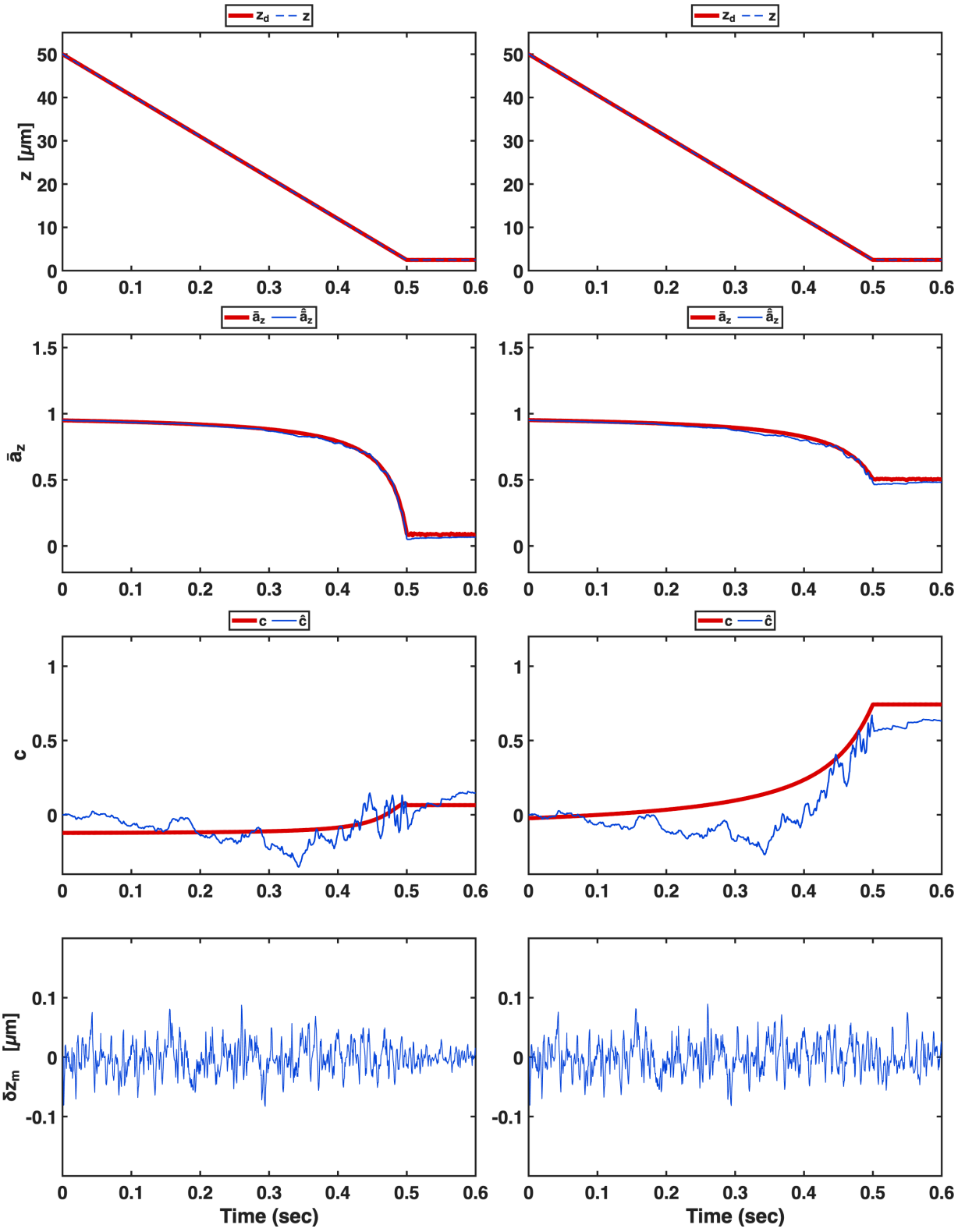


Fig. 6.5 Wall type read from $\hat{c}$ at the end of the descent

z axis; 22.2 → 1.10 R in 0.5 s, then hold 0.1 s; $\sqrt{P_{\text{fix}}} = 0.15$; left: plane wall; right: cell wall; $\lambda_c = 0.4$, seed 1



7 Reference result: raster scan parallel to the wall

Fig. 7.1 Raster scan: scan pattern and trajectory

left: y vs x, desired (red) and actual (blue dashed); right: x, y, z vs time, desired (red) and actual (blue dashed); x_d 4 Hz triangle ± 20 μm, y_d $+1$ μm at every x turnaround, z_d held at 1.10 R; plane wall, $\lambda_c = 0.4$, seed 1

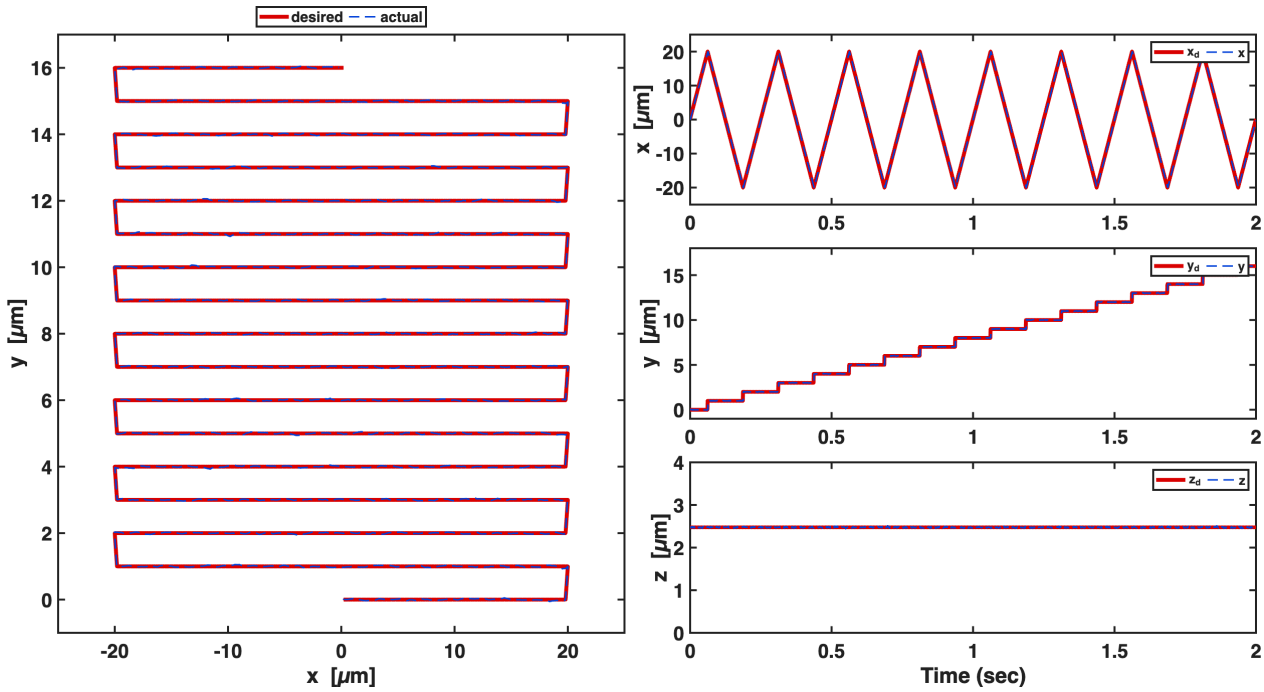
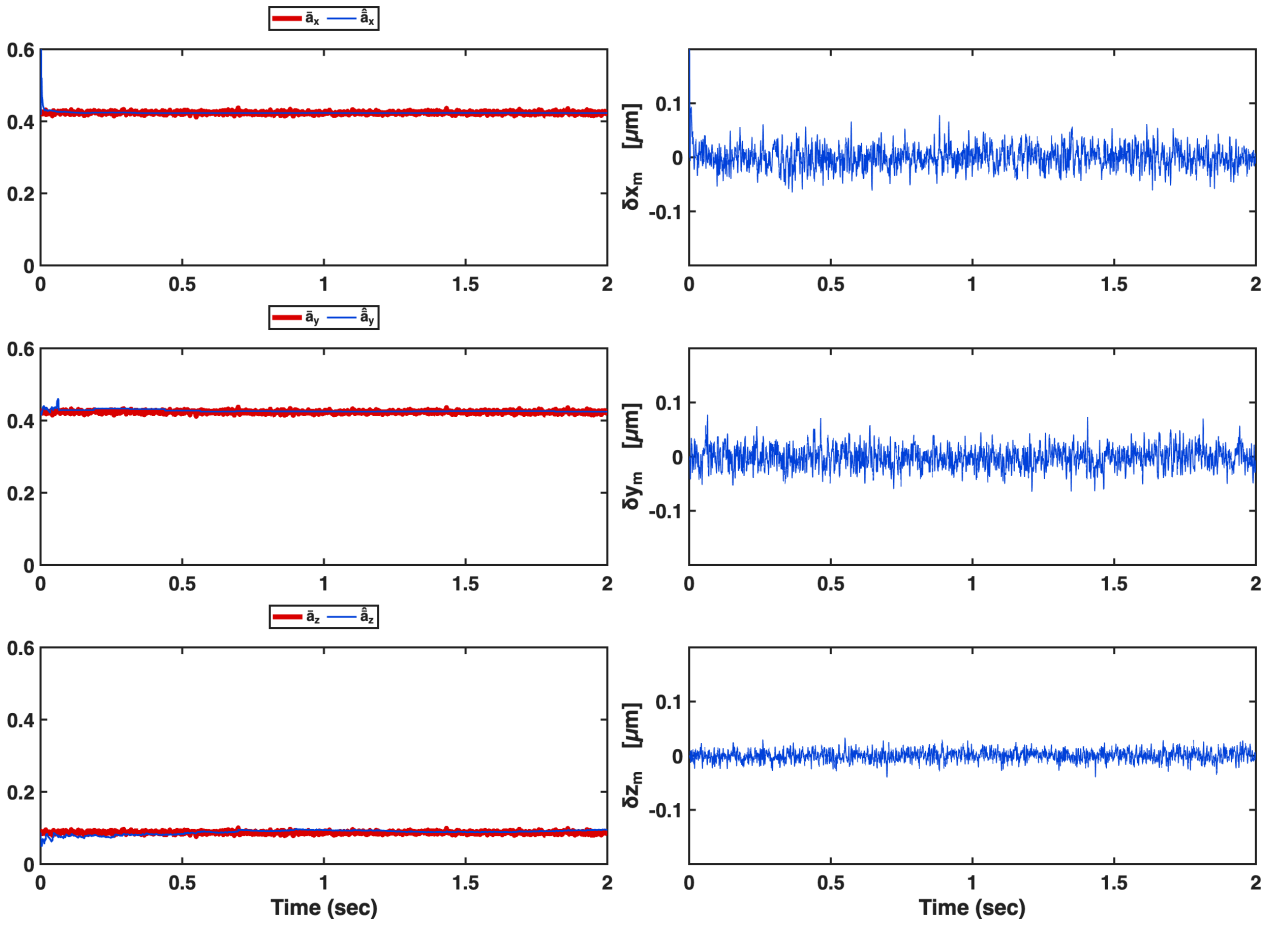


Fig. 7.2 Raster scan: motion gain and measured tracking error, three axes

left: true (red) and estimated (blue) motion gain; right: measured tracking error; z: Section 6 estimator; x, y: plane parallel law driven by the wall-normal step; plane wall, $\lambda_c = 0.4$, seed 1



8 Slope with unit numerator, the wall position as the state

Motion gain as a function of the distance from the wall

$$
\begin{aligned}
h[k] &= \bar{x}_d[k] - \delta\bar{x}_3[k] - \bar{w}_s \\
\bar{a}'[k] &= \frac{1}{h[k]^2} \\
M[k] &= \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\bar{x}_3[k]
\end{aligned}
$$

State of the process

$$
\begin{aligned}
\delta\bar{x}_1[k+1] &= \delta\bar{x}_2[k] \\
\delta\bar{x}_2[k+1] &= \delta\bar{x}_3[k] \\
\delta\bar{x}_3[k+1] &= \lambda_c\,\delta\bar{x}_3[k] - F_{dx}[k]\,e_{\bar{a}}[k] - \varepsilon_x[k] \\
\bar{a}[k+1] &= \bar{a}[k] + \bar{a}'[k]\left(M[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\right) \\
\bar{w}_s[k+1] &= \bar{w}_s[k]
\end{aligned}
$$

Measurement and output matrix

$$
\mathbf{y}[k] = \begin{bmatrix} \delta\bar{x}_m[k] \\ \bar{a}_m[k] \end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix} \delta\hat{\bar{x}}_1[k] \\ \hat{\bar{a}}[k] \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}
$$

Estimator

$$
\begin{aligned}
\hat{h}[k] &= \bar{x}_d[k] - \delta\hat{\bar{x}}_3[k] - \hat{\bar{w}}_s[k] \\
\hat{\bar{a}}'[k] &= \frac{1}{\hat{h}[k]^2} \\
\hat{M}[k] &= \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{x}}_3[k]
\end{aligned}
$$

$$
\begin{aligned}
\delta\hat{\bar{x}}_1[k+1] &= \delta\hat{\bar{x}}_2[k] + \left\{\ell_{11}[k]\,e_{x_1}[k] + \ell_{12}[k]\,e_{a_m}[k]\right\} \\
\delta\hat{\bar{x}}_2[k+1] &= \delta\hat{\bar{x}}_3[k] + \left\{\ell_{21}[k]\,e_{x_1}[k] + \ell_{22}[k]\,e_{a_m}[k]\right\} \\
\delta\hat{\bar{x}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{x}}_3[k] + \left\{\ell_{31}[k]\,e_{x_1}[k] + \ell_{32}[k]\,e_{a_m}[k]\right\} \\
\hat{\bar{a}}[k+1] &= \hat{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\hat{M}[k] + \left\{\ell_{41}[k]\,e_{x_1}[k] + \ell_{42}[k]\,e_{a_m}[k]\right\} \\
\hat{\bar{w}}_s[k+1] &= \hat{\bar{w}}_s[k] + \left\{\ell_{51}[k]\,e_{x_1}[k] + \ell_{52}[k]\,e_{a_m}[k]\right\}
\end{aligned}
$$

Dynamics of estimation errors

$$
\mathbf{e}[k] = \begin{bmatrix} e_{\delta\bar{x}_1}[k] \\ e_{\delta\bar{x}_2}[k] \\ e_{\delta\bar{x}_3}[k] \\ e_{\bar{a}}[k] \\ e_{\bar{w}_s}[k] \end{bmatrix}, \qquad \mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k]
$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & 0 \\ 0 & 0 & \hat{\bar{a}}'[k]\,(1-\lambda_c) + \dfrac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k] & 1 + \hat{\bar{a}}'[k]\,F_{dx}[k] & \dfrac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k] \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{q}[k] = \begin{bmatrix} 0 \\ 0 \\ -\varepsilon_x[k] \\ \hat{\bar{a}}'[k]\,\varepsilon_x[k] \\ 0 \end{bmatrix}$$

Row 4 of $\mathbf{F}_e$, expanded

(a) the two gain steps and their difference

$$\bar{a}[k+1] - \bar{a}[k] = \bar{a}'[k]\,\big(M[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\big)$$
$$\hat{\bar{a}}[k+1] - \hat{\bar{a}}[k] = \hat{\bar{a}}'[k]\,\hat{M}[k]$$
$$e_{\bar{a}}[k+1] - e_{\bar{a}}[k] = \bar{a}'[k]\,M[k] - \hat{\bar{a}}'[k]\,\hat{M}[k] + \bar{a}'[k]\,F_{dx}[k]\,e_{\bar{a}}[k] + \bar{a}'[k]\,\varepsilon_x[k]$$

(b) each factor written as its estimate plus its error

$$h[k] = \hat{h}[k] - e_{\delta\bar{x}_3}[k] - e_{\bar{w}_s}[k]$$
$$\bar{a}'[k] = \frac{1}{\big(\hat{h}[k] - e_{\delta\bar{x}_3}[k] - e_{\bar{w}_s}[k]\big)^2} = \hat{\bar{a}}'[k] + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\big(e_{\delta\bar{x}_3}[k] + e_{\bar{w}_s}[k]\big) + O(e^2)$$
$$M[k] = \hat{M}[k] + (1-\lambda_c)\,e_{\delta\bar{x}_3}[k]$$

(c) the product difference, exact

$$\bar{a}'[k]\,M[k] - \hat{\bar{a}}'[k]\,\hat{M}[k] = \hat{\bar{a}}'[k]\,\big(M[k] - \hat{M}[k]\big) + \big(\bar{a}'[k] - \hat{\bar{a}}'[k]\big)\,\hat{M}[k] + \big(\bar{a}'[k] - \hat{\bar{a}}'[k]\big)\big(M[k] - \hat{M}[k]\big)$$

(d) every term after substituting (b)

$$\begin{aligned} e_{\bar{a}}[k+1] - e_{\bar{a}}[k] = {} & \hat{\bar{a}}'[k]\,(1-\lambda_c)\,e_{\delta\bar{x}_3}[k] \\ & + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\delta\bar{x}_3}[k] + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\bar{w}_s}[k] \\ & + \hat{\bar{a}}'[k]\,F_{dx}[k]\,e_{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\varepsilon_x[k] \\ & + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\big(e_{\delta\bar{x}_3}[k] + e_{\bar{w}_s}[k]\big)\Big((1-\lambda_c)\,e_{\delta\bar{x}_3}[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\Big) + O(e^2) \end{aligned}$$

(e) first-order terms collected by error

$$\begin{aligned} e_{\bar{a}}[k+1] = {} & \Big(\hat{\bar{a}}'[k]\,(1-\lambda_c) + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\Big)\,e_{\delta\bar{x}_3}[k] \\ & + \big(1 + \hat{\bar{a}}'[k]\,F_{dx}[k]\big)\,e_{\bar{a}}[k] \\ & + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\bar{w}_s}[k] \\ & + \hat{\bar{a}}'[k]\,\varepsilon_x[k] \end{aligned}$$

Process noise and measurement noise

$$\mathbf{Q}[k] = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & q_{33}[k] & -\hat{\bar{a}}'[k]\,q_{33}[k] & 0 \\ 0 & 0 & -\hat{\bar{a}}'[k]\,q_{33}[k] & \hat{\bar{a}}'[k]^2\,q_{33}[k] & 0 \\ 0 & 0 & 0 & 0 & q_{55} \end{bmatrix}, \qquad q_{55} = 0$$

$$\mathbf{R}[k] = \mathrm{diag}\big(r_1,\ r_2[k]\big)$$

(q_{33}, r_1, r_2, $\mathbf{L}$, $\mathbf{P}$, $\mathbf{P}^f$ as in Section 4.)

Fig. 8.1 Estimator started from a wrong wall position: plane and cell walls

z axis; 22.2 → 1.10 R in 1 s, then hold 0.2 s; $\hat{\bar{w}}_s[0] = -1$ R, $\sqrt{p_{55}[0]} = 0.3$ R, $q_{55} = 0$; left: plane wall; right: cell wall (plane polynomial slid by 1.05 R, plant only); $\bar{w}_s$ row: red = equivalent wall, the position at which the plane law reproduces the wall (plane 0, cell −1.05 R), grey = visible surface (cell), dashed = $\bar{w}_s$ that makes $1/(h - \bar{w}_s)^2$ equal the true slope at the true height, band = $\pm\sqrt{p_{55}}$; $\lambda_c = 0.4$, seed 1

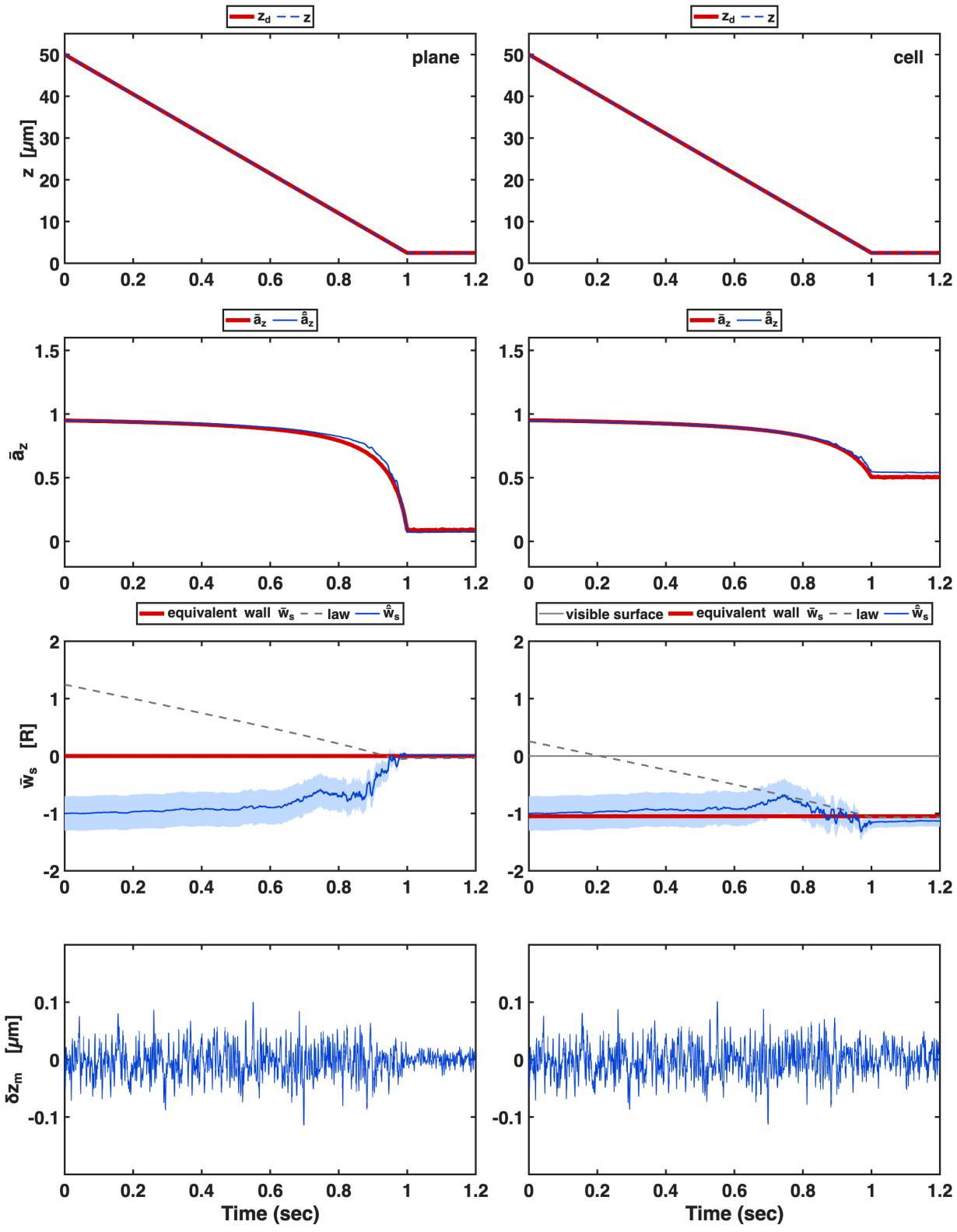
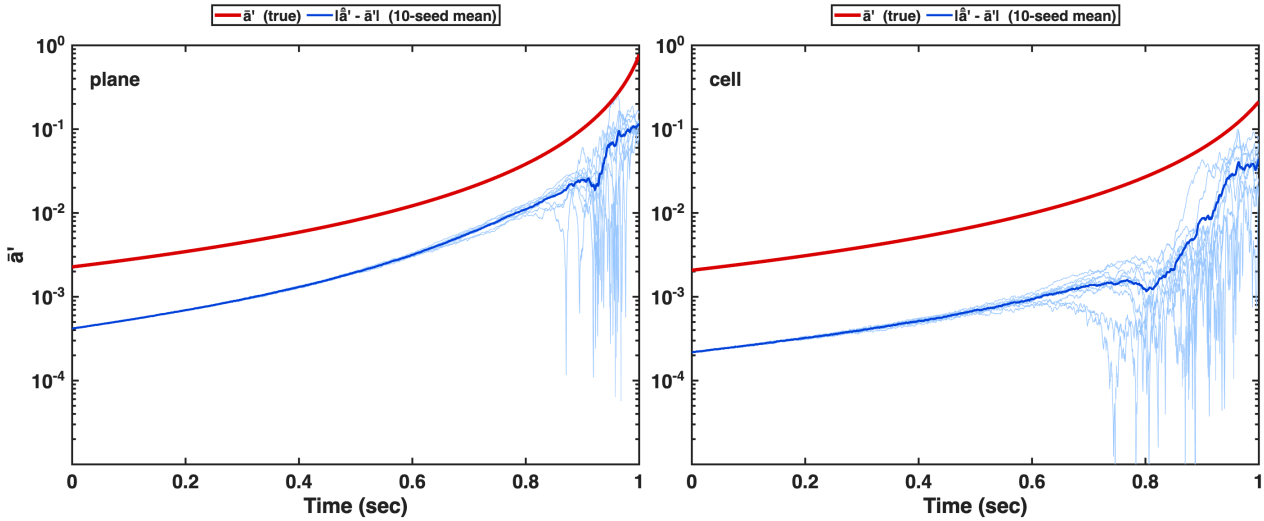


Fig. 8.2 Error of the slope along the descent

run of Fig. 8.1, 10 seeds (light: each seed, dark: mean); $|\hat{\bar{a}}' - \bar{a}'|$ with $\hat{\bar{a}}' = 1/(h_d - \hat{\bar{w}}_s)^2$ and $\bar{a}'$ the slope of the plant at the true height (red); left: plane wall; right: cell wall; descent only



9 Regulation of the motion gain

Gain step of the probe (Section 8)

$$\bar{a}[k+1] - \bar{a}[k] = \bar{a}'[k]\left(\Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\bar{x}_3[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\right)$$

Regulation objective and law

$$\begin{aligned}
\bar{a}_d - \bar{a}[k+1] &= \lambda_a\left(\bar{a}_d - \bar{a}[k]\right) \\
\Delta\bar{x}_d[k] &= (1-\lambda_a)\,\frac{\bar{a}_d - \hat{\bar{a}}[k]}{\hat{\bar{a}}'[k]} \\
\bar{x}_d[k+1] &= \bar{x}_d[k] + \Delta\bar{x}_d[k] \\
\lambda_a &= 1 - T_s/\tau_a, \qquad \tau_a = 0.1\,\mathrm{s}
\end{aligned}$$

Fig. 9.1 Gain regulation to $\bar{a}_d = 0.07$ on the plane wall

z axis; start parked at 22.2 R, run 6 s; $\tau_a = 0.1$ s, $\hat{\bar{w}}_s[0] = 0$; estimator of Section 8 ($\sqrt{p_{55}[0]} = 0.3$ R, $q_{55} = 0$); rows: height (log scale), gain (red true, blue estimate, dashed $\bar{a}_d$), wall position (red = equivalent wall, the position at which the plane law reproduces the wall (plane 0, cell -1.05 R), grey = visible surface, dashed = $\bar{w}_s$ that makes $1/(h - \bar{w}_s)^2$ equal the true slope at the true height), δz_m; plant gain $\bar{a}_d = 0.07$ at $h = 1.078$ R (gap 176 nm); $\lambda_c = 0.4$, seed 1

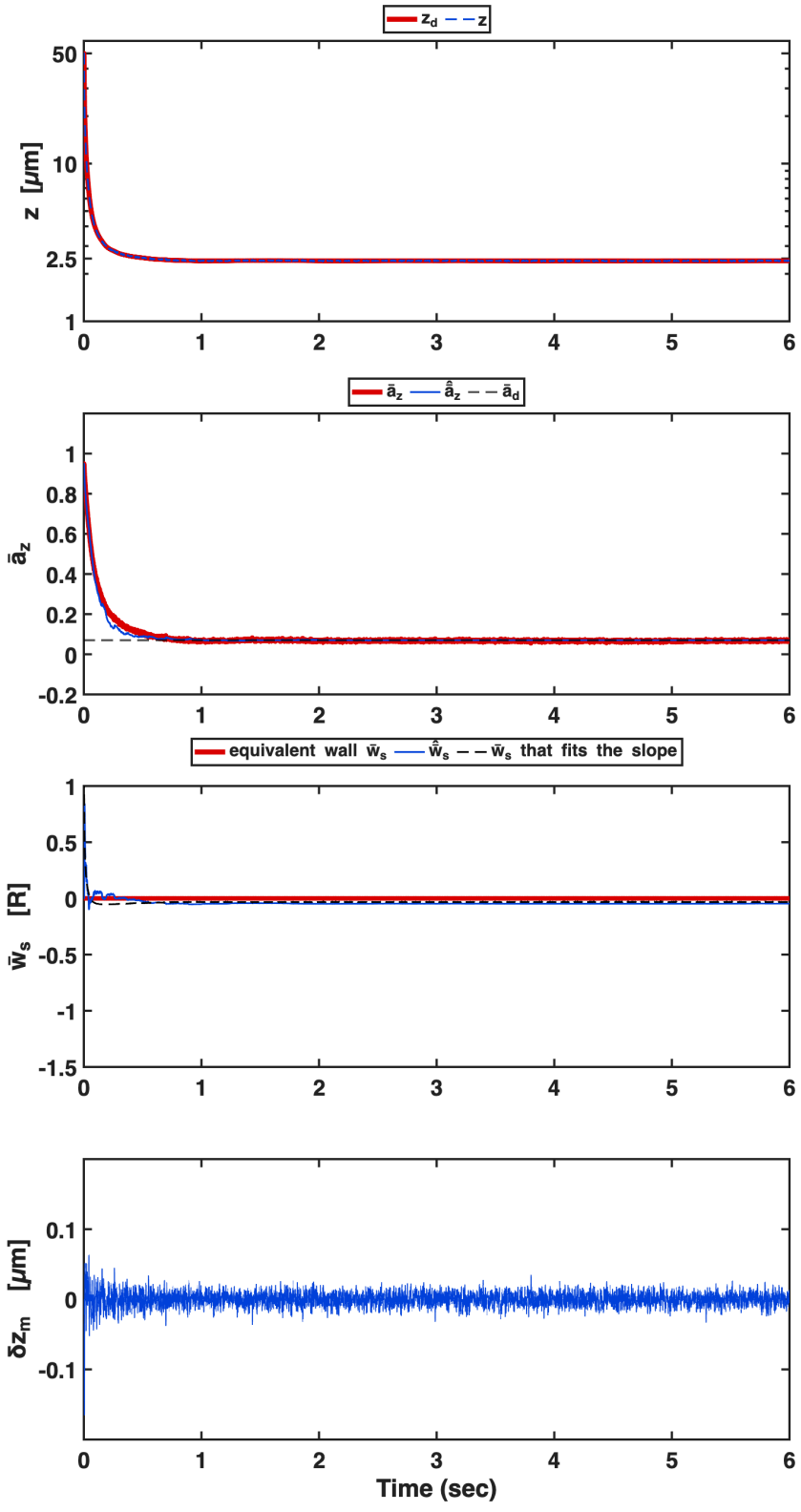


Fig. 9.2 Wrong start of the wall position: plane wall, $\bar{a}_d = 0.07$

z axis; start parked at 22.2 R, run 6 s; $\tau_a = 0.1$ s; columns: $\hat{\bar{w}}_s[0] = -3$, 0, +3 R; estimator of Section 8 ($\sqrt{p_{55}[0]} = 0.3$ R, $q_{55} = 0$); rows: gain (log scale; red true, blue estimate, dashed $\bar{a}_d$), wall position (red = equivalent wall, the position at which the plane law reproduces the wall (plane 0, cell −1.05 R), grey = visible surface, blue estimate, dashed = $\bar{w}_s$ that makes $1/(h - \bar{w}_s)^2$ equal the true slope at the true height); plant gain $\bar{a}_d = 0.07$ at $h = 1.078$ R (gap 176 nm); $\lambda_c = 0.4$, seed 1

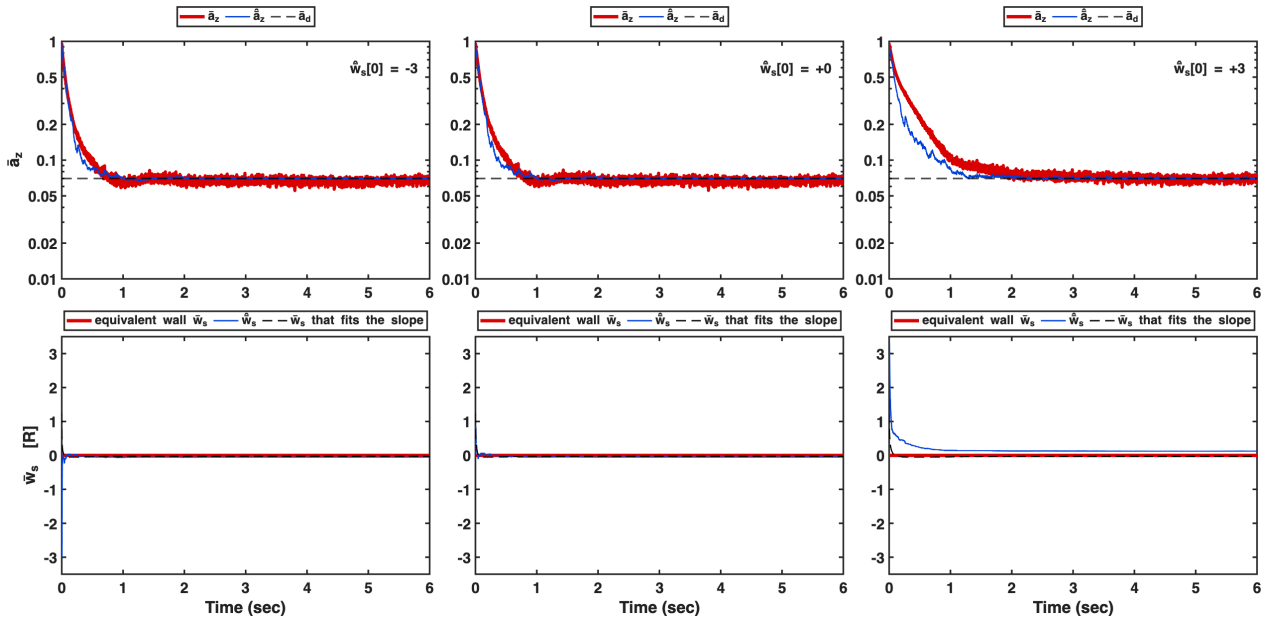


Fig. 9.3 Gain regulation to $\bar{a}_d = 0.5$ on the plane wall

z axis; start parked at 22.2 R, run 6 s; $\tau_a = 0.1$ s, $\hat{\bar{w}}_s[0] = 0$; estimator of Section 8 ($\sqrt{p_{55}[0]} = 0.3$ R, $q_{55} = 0$); rows: height (log scale), gain (red true, blue estimate, dashed $\bar{a}_d$), wall position (red = equivalent wall, the position at which the plane law reproduces the wall (plane 0, cell -1.05 R), grey = visible surface, dashed = $\bar{w}_s$ that makes $1/(h - \bar{w}_s)^2$ equal the true slope at the true height), δz_m; plant gain $\bar{a}_d = 0.5$ at $h = 2.128$ R (gap 2538 nm); $\lambda_c = 0.4$, seed 1

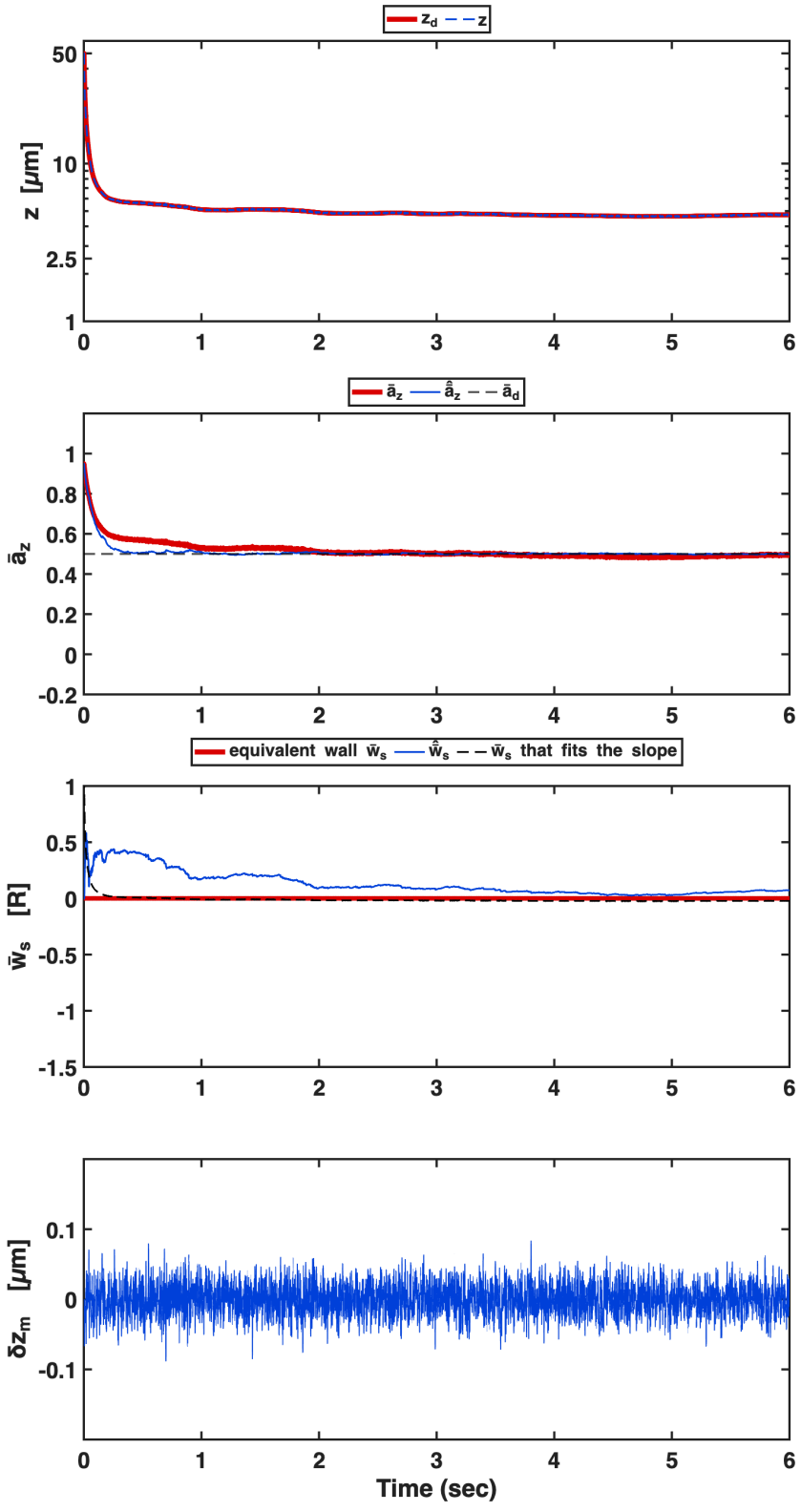


Fig. 9.4 Gain regulation to $\bar{a}_d = 0.50$ on the cell wall

z axis; start parked at 22.2 R, run 6 s; $\tau_a = 0.1\,\mathrm{s}$, $\hat{\bar{w}}_s[0] = 0$ (estimator started from the plane); cell wall = plane polynomial slid by 1.05 R, plant only; estimator of Section 8 ($\sqrt{p_{55}[0]} = 0.3$ R, $q_{55} = 0$); rows: height (log scale), gain (red true, blue estimate, dashed $\bar{a}_d$), wall position (red = equivalent wall, the position at which the plane law reproduces the wall (plane 0, cell -1.05 R), grey = visible surface, dashed = $\bar{w}_s$ that makes $1/(h - \bar{w}_s)^2$ equal the true slope at the true height), δz_m; plant gain $\bar{a}_d = 0.50$ at $h = 1.078$ R (gap 175 nm from the visible surface); $\lambda_c = 0.4$, seed 1

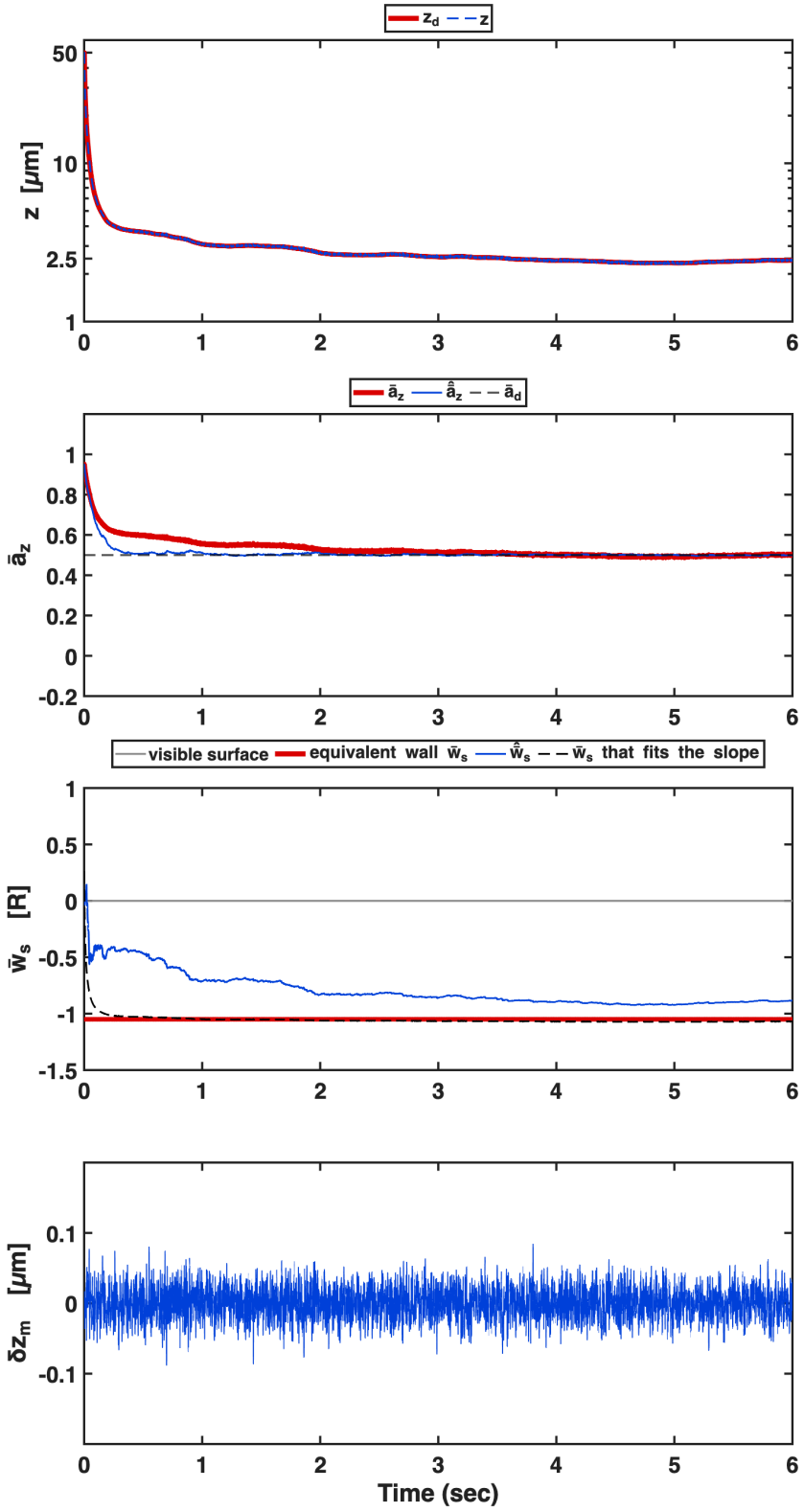

z_d z
z [μm]
50
10
2.5
1
0 1 2 3 4 5 6
$\bar{a}_z$ $\hat{\bar{a}}_z$ $\bar{a}_d$
$\bar{a}_z$
1
0.8
0.6
0.4
0.2
0
-0.2
0 1 2 3 4 5 6
visible surface
equivalent wall $\bar{w}_s$
$\hat{\bar{w}}_s$
$\bar{w}_s$ that fits the slope
$\bar{w}_s$ [R]
1
0.5
0
-0.5
-1
-1.5
0 1 2 3 4 5 6
δz_m [μm]
0.1
0
-0.1
0 1 2 3 4 5 6
Time (sec)

Fig. 9.5 Wrong start of the wall position: cell wall, $\bar{a}_d = 0.52$

z axis; start parked at 22.2 R, run 6 s; $\tau_a = 0.1$ s; columns: $\hat{\bar{w}}_s[0] = -3$, 0, +3 R; cell wall = plane polynomial slid by 1.05 R, plant only; estimator of Section 8 ($\sqrt{p_{55}[0]} = 0.3$ R, $q_{55} = 0$); rows: gain (log scale; red true, blue estimate, dashed $\bar{a}_d$), wall position (red = equivalent wall, the position at which the plane law reproduces the wall (plane 0, cell -1.05 R), grey = visible surface, blue estimate, dashed = $\bar{w}_s$ that makes $1/(h - \bar{w}_s)^2$ equal the true slope at the true height); plant gain $\bar{a}_d = 0.52$ at $h = 1.174$ R (gap 392 nm from the visible surface); $\lambda_c = 0.4$, seed 1

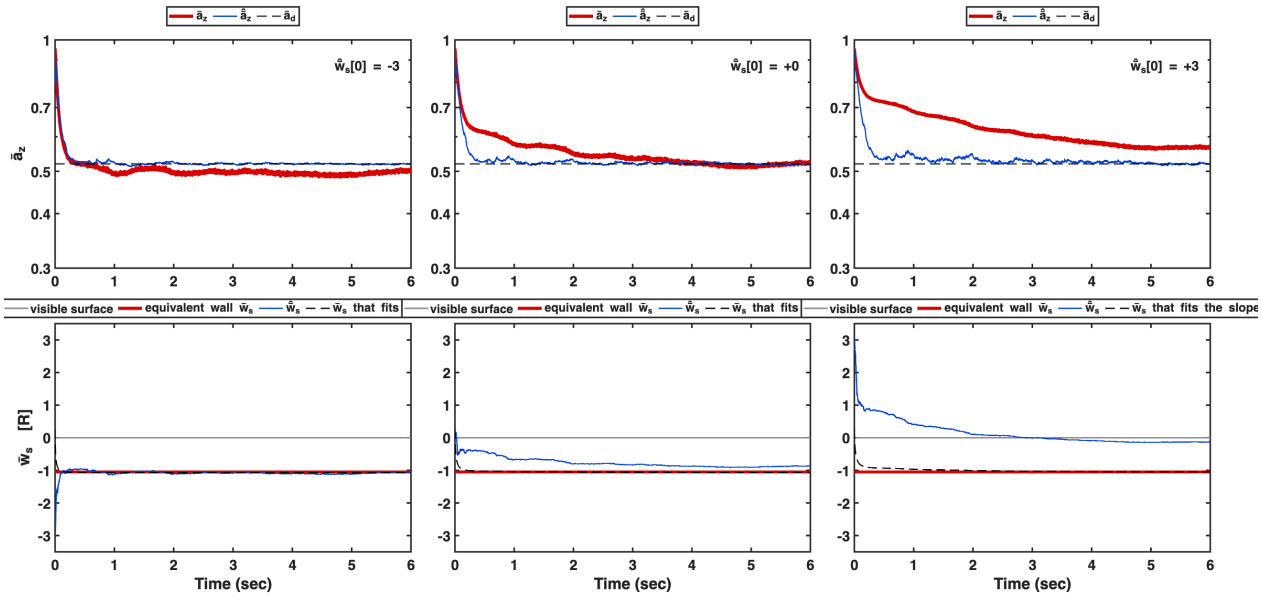


Fig. 9.6 Wall position alone vs wall position and exponent: cell wall, $\bar{a}_d = 0.52$

z axis; start parked at 22.2 R, run 6 s; $\tau_a = 0.1\,\mathrm{s}$, $\hat{\bar{w}}_s[0] = 0$; cell wall = plane polynomial slid by 1.05 R, plant only; left: $\hat{\bar{a}}' = 1/(h - \hat{\bar{w}}_s)^2$; right: $\hat{\bar{a}}' = 1/(h - \hat{\bar{w}}_s)^{\hat{p}}$ ($\hat{p}[0] = 2$, $\sqrt{p_{pp}[0]} = 0.1$); $\sqrt{p_{55}[0]} = 0.3$ R, $q = 0$; rows: gain (red true, blue estimate, dashed $\bar{a}_d$), wall position (red = equivalent wall, the position at which the plane law reproduces the wall (plane 0, cell -1.05 R), grey = visible surface, blue estimate), exponent (left: fixed 2; right: $\hat{p}$ blue, dashed $p_{\mathrm{eff}}(h)$); plant gain $\bar{a}_d = 0.52$ at $h = 1.174$ R (gap 392 nm from the visible surface); 10 seeds, paired (p estimated minus $p = 2$): mean $|\bar{a} - \bar{a}_d|$ -0.007 ± 0.002 (1–2 s), -0.004 ± 0.002 (2–4 s), $+0.002 \pm 0.001$ (4–6 s); end distance of $\hat{\bar{w}}_s$ to the physical wall 0.134 → 0.081 R (-0.053 ± 0.014, closer in 8 of 10 seeds); $\hat{p}$ ends at 1.986 ± 0.008: freeing the exponent moves the wall estimate toward the physical wall and speeds the first seconds, but does not improve the regulated gain at the end; plotted: seed 1; $\lambda_c = 0.4$

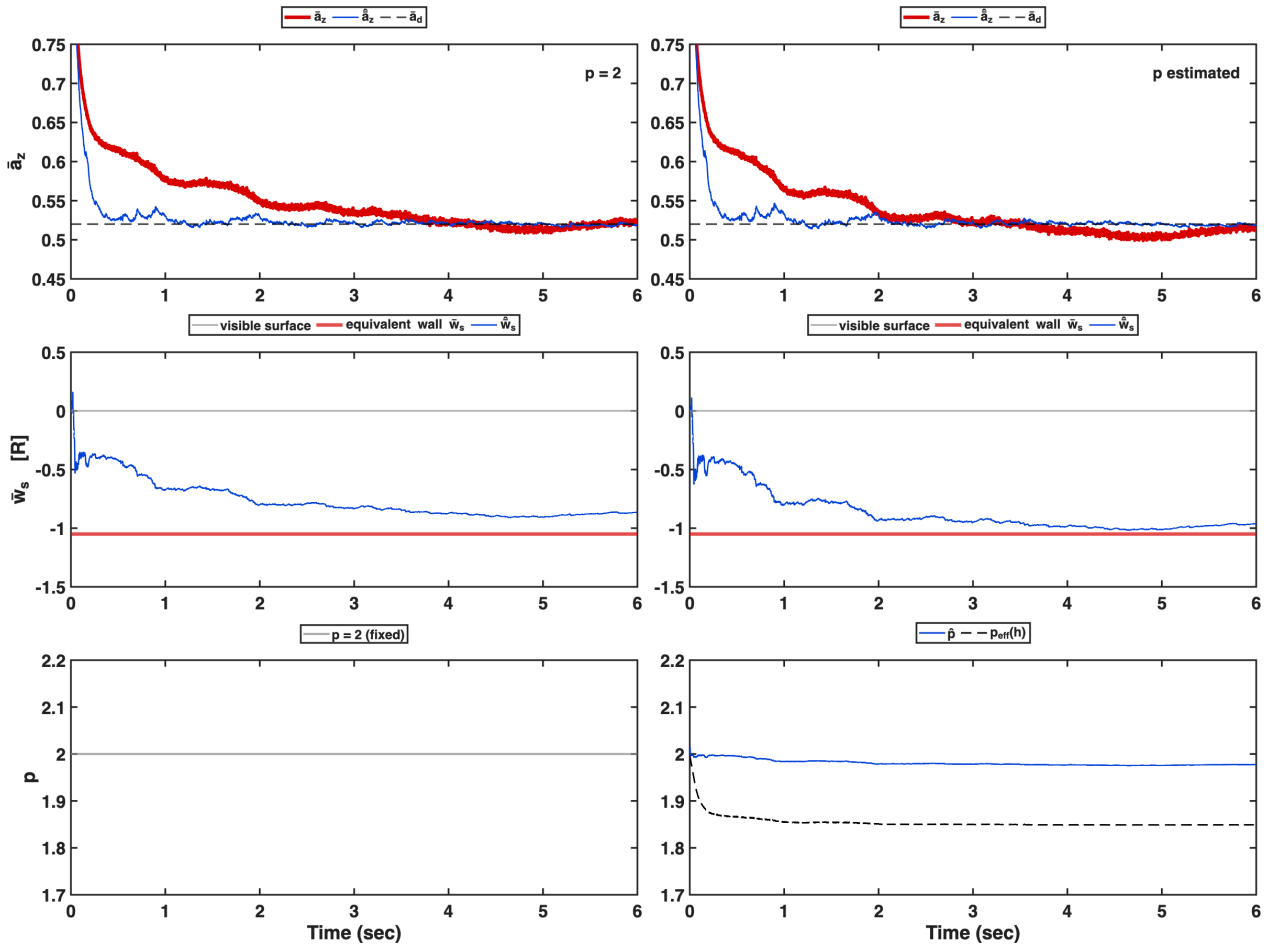


Fig. 9.7 Estimator started at the assumed wall, true wall moved: plane wall, $\bar{a}_d = 0.07$

z axis; start parked at 22.2 R, run 6 s; $\tau_a = 0.1$ s; estimator of Section 8 always started at $\hat{\bar{w}}_s[0] = 0$ ($\sqrt{p_{55}[0]} = 0.3$ R); the true wall is moved by Δ on the plant side only ($\Delta > 0$: closer to the probe than assumed); columns $\Delta = -2, -1, +2$ R; rows: true gap h (blue; red = command in the true frame; dashed $h^* = 1.078$ R where the plant gain equals $\bar{a}_d$), gain (red true, blue estimate, dashed $\bar{a}_d$), wall position (red = equivalent wall in the estimator frame, Δ on the plane and $\Delta - 1.05$ R on the cell, grey = visible surface Δ, blue estimate); 5 seeds, last 0.5 s, mean (worst seed) of $h - h^*$ and $\bar{a} - \bar{a}_d$: $\Delta = +3, +2, +1, 0$: -0.003 (-0.006) R and -0.002 (-0.005), $\hat{\bar{w}}_s$ within 0.05 R of Δ; $\Delta = -1$: $+0.015$ ($+0.048$) R and $+0.012$ ($+0.037$), $\hat{\bar{w}}_s = -1.02$; $\Delta = -2$: every seed touches the wall (seed 1 at 2.5 s, after $\hat{\bar{w}}_s$ had reached -1.9 while $\bar{a}$ stayed at 0.14); $\Delta = -3$: some seeds touch the wall; larger positive Δ: plane works through $+5$ (the range limit of $\hat{\bar{w}}_s$; $+6$ needs the limit widened), $+8$ and $+10$ touch the wall; cell wall at $\bar{a}_d = 0.52$ ($h^* = 1.174$ R): $\Delta = 0$ $+0.005$ (-0.065) R, $+1$ -0.037 (-0.102) R, -1 $+0.436$ ($+0.705$) R, $+2$ and above touch the wall; $\lambda_c = 0.4$, plotted seed 1

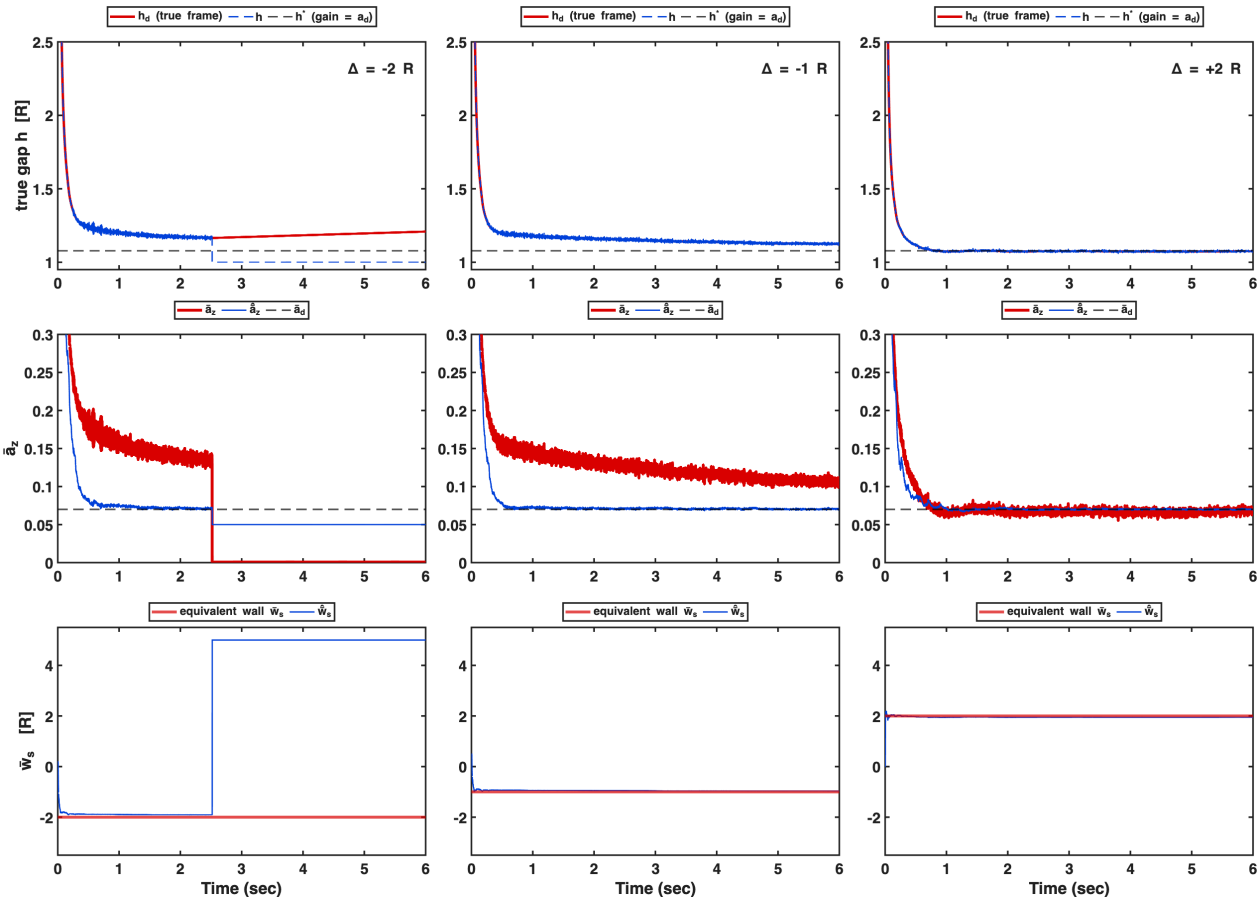


Fig. 9.8 Estimator started at the assumed wall, true wall moved: cell wall, $\bar{a}_d = 0.52$

z axis; start parked at 22.2 R, run 6 s; $\tau_a = 0.1$ s; estimator of Section 8 always started at $\hat{\dot{w}}_s[0] = 0$ ($\sqrt{p_{55}[0]} = 0.3$ R); the true wall is moved by Δ on the plant side only; cell wall = plane polynomial slid by 1.05 R, so the true wall sits at $\Delta - 1.05$ R in the estimator frame; columns $\Delta = -1$, 0, +1 R; rows as Fig. 9.7 ($h^* = 1.174$ R, gap 392 nm from the visible surface); 5 seeds, last 0.5 s, mean (worst seed) of $h - h^*$: $\Delta = -1$ +0.436 (+0.705) R, 0 +0.005 (−0.065) R, +1 −0.037 (−0.102) R; $\Delta = +2$ and above touch the wall; $\lambda_c = 0.4$, plotted seed 1

